ZAZA Pro

1 Meta Information

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Addressees and Intention of this Document	This document is intended to describe the product by **summarizing all important product definitions**, such as **high-level system architecture, intended use, action and function principle, clinical benefit** and **overall risk considerations** as well as technical data and a photo documentation. This document is intended for all stakeholders to receive a comprehensive overview.
Changes since last version	Defined product description definition in a more specific way Mentioned CDS Tool AT01 'ESC 2016 Atrial Fibrillation Dashboard' and its tool description and intended use Added Chapter 3.5 Product Labelling
Documents needed (Relevant Documents)	TD_0100_001_Product History CDS Tool List.pdf
Definitions / Abbreviations	Refer to Chapter 6.3 Glossary

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2 Introduction to Clinical Decision Support

2.1 Data Model of Medicine

Medicine is the science and practice of the diagnosis, treatment, and prevention of diseases. Practicing medicine means **providing the most suitable care to a patient**, thereby matching the physician's personal medical knowledge (data) against the presenting data of the patient (= knowledge/data coupling), then making decisions (ruling, reasoning) based on these data and performing indicated activities. Medicine is thinking and acting on patient data [6]. Providing healthcare to a patient means **managing patient problems**, using evidence-based, predefined problem solution strategies (schemes).

2.2 Guideline Statements

Clinical guidelines are systematically developed statements to assist practitioners with decision making about appropriate health care for specific clinical circumstances (Field and Lohr, 1990). Clinical guidelines are typically created by medical experts or panels convened by specialty organizations, who review the relevant studies, perform meta-analyses by contrasting and combining the results from different studies and, using a consensus-based process, compile a set of evidence-based recommendations [2].

Guidelines help clinicians translate best evidence into best practice. A well-crafted guideline promotes quality by reducing health care variations, improving diagnostic accuracy, promoting effective therapy, and discouraging ineffective - or potentially harmful - interventions [4]. Guidelines prescribe **recommendations in key action statements** followed by supporting (amplifying) text.

Example of a guideline action statement: “Other pharmacological treatments recommended in selected patients with symptomatic (NYHA Class II-IV) heart failure with reduced ejection fraction: Diuretics are recommended in order to improve symptoms and exercise capacity in patients with signs and/or symptoms of congestion.” (European Society of Cardiology, 2016)

All guideline action statements should ideally be supported by an **action statement profile** that summarizes clearly the decision-making process in terms of aggregate evidence quality, harm-benefit assessment, development group values, reasons for intentional vagueness, and the role of patient preference [4].

2.3 Clinical Decision Support Systems

Computer-based clinical decision support (CDS) can be defined as the use of information and communication technologies to bring relevant knowledge to bear on the health care and well-being of a patient [2]. A **clinical decision support system (CDSS)** is understood here as a software system designed to apply automated reasoning on **digitized patient data** following **provided clinical rules** and to demonstrate **recommendations to users** (physicians) for the diagnosis, prevention, monitoring, treatment or alleviation of a disease. A **clinical decision** is meant to be ruling the management of health and healthcare of an individual person (patient) [2]. **Support** is meant to be the **aiding of**, rather than the making of decisions [2], which means demonstrating recommendations to healthcare professionals.

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There is always an **intermediary**, the recipient (user) in the loop between the recommendation of an activity/decision and the real activity/decision. Clinical decision **support** is therefore not a "closed-loop" process. CDS systems improve quality, efficiency, transparency and safety of healthcare [2].

2.4 Mathematical Concepts Used in CDS Systems

Essential to the understanding of clinical decision support systems (CDSS) is familiarity with the basic principles of logic and probability [1]. Mathematical concepts are key to organize medicine and medical data within software systems and to make clinical problems and solutions machine-readable, machine-understandable and machine-processable. CDS systems are based on the application of the mathematical concepts of **set theory** (Mengenlehre), **Boolean algebra**, **probability theory** (stochastics), **fuzzy set theory** and **decision theory**.

Set theory is a branch of mathematical logic that studies sets. A **set** is a well-defined collection of unique objects or elements (e.g. a set of polygons). Sets are used to **formalize logical operations** in a way that machines (computers) can understand. **Boolean algebra** is a formalism for describing logical relations in the same way that ordinary algebra describes numeric relations. Boolean algebra has been fundamental in the development of digital electronics. Computerized systems can use some numerical representation of **probability** in order to combine statements into conclusions. **Fuzzy logic** is used to handle decision making problems in which uncertainty plays a role. A decision rule in medicine is a representation of **decision theory** in which branching logic is used in combination with data to reach conclusions regarding diagnosis, treatment and other important clinical goals.

2.5 Clinical Decision Rules

2.5.1 Structure of Clinical Decision Rules

A physician is making his clinical decisions based on **clinical decision rules** which are part of his personal medical knowledge. A **clinical decision support system (CDSS)** is supposed to aid these decision-making processes by presenting recommendations based on **clinical decision rules** that were earlier determined and stored within the CDSS knowledge base. These clinical decision rules are also called **production rules** or **condition-action-rules**. A clinical decision rule maps from the characteristics of the situation to the behavior that should be performed or the conclusions that should be reached in that situation.

The conventional format for a production rule is the **IF-THEN statement**:
IF <condition> THEN <action>
where <condition> represents a logical statement that, if true, leads to the <action> being undertaken. The **condition part** is sometimes known as the **left-hand side (LHS)** of the statement, while the **action part** is known as the **right-hand side (RHS).**

The condition may be a simple, single comparison involving data available to the CDSS, or it may be an arbitrarily complex statement in Boolean logic, using association, conjunction, disjunction and negation related to data. The action or RHS of a production rule may be an instruction to generate a message, usually a recommendation that some action will be performed by a person, or a conclusion, typically represented by an assignment statement

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(order) that contributes another fact or data element available to the CDSS. In addition, like the LHS of a rule, the RHS may be arbitrarily complex and consist of several actions.
A **knowledge base** represented using production rules would consist of a collection of these condition action statements. The **data elements** against which the knowledge base would apply would consist of **data elements** that describe the **conditions** of a patient.

2.5.2 Clinical Decision Rule Profiles

Just as guideline action statements should ideally be supported by a **guideline action statement profile**, clinical decision rules should be backed up by a **clinical decision rule profile**.

For transparency and legal reasons, a decision rule profile should clearly summarize the underlying guideline action statements or alternative decision-making processes e.g. a risk-benefit assessment. A **clinical decision rule profile** should (at least) list the following items:

- Complete **original clinical decision rule**, the statement text (recommendation) in an analog form, e.g. as taken out of a clinical practice guideline text
- **Original source** and author of the statement, e.g. a medical society
- **Digitized by** and reviewed by who
- **Indications for use**, that is who and what the rule applies to: **patient group**, anatomical structure, **disease** or injury
- **Clinical benefits, clinical risks and risk-benefit conclusion**

When a clinical decision rule is based on a guideline action statement, the **guideline action statement profile** can - if available – partially be used as a **clinical decision rule profile**. If no guideline action statement profile is available, other information from the guideline, like the guideline statement grading (level of evidence, grade of recommendation) should be listed. If no guideline is available for the problem at all or local rules are to be adhered to, a complete clinical decision rule profile can alternatively be filled out by an admin user.

2.6 Problem-based Clinical Decision Support

2.6.1 Clinical Patient Problems

Providing healthcare to a patient means **managing patient problems**, using evidence-based, predefined problem solution strategies (schemes). **Clinical patient problems are "what is wrong with the patient"**. Patient problems can be described by medical data (elements).

2.6.2 The Problem List

Each medical record should have a complete list of all the patients' problems, including both **clearly established diagnoses** and all other **unexplained findings** that are not yet clear manifestations of a specific diagnosis, such as abnormal physical findings or symptoms. The problems list then is not static in its composition, but is a dynamic table of contents of the patient's chart, which can be updated at any time [7]. Anything important in the database goes into the problem list: signs, symptoms, social burdens, previously diagnosed disorders, abnormal lab and X-ray findings. **Active problems** need to be distinguished from **inactive** problems. Active problems require attention. Resolved problems can interfere with the current management and should therefore be born in mind. The **dates when problems**

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started should be noted (month, year). The problem list is a dynamic index, that is continuously updated [5].

To receive a problem list for each patient, the complete history should be taken first. A full physical exam should be done. Based on the history and physical examination a database needs to be written down. Then the problem list is compiled. The problems can come from any part of the database. Later in the process of treating a patient unexplained finding of the problem list can become part of one single established diagnosis [5].

2.6.3 Problem List and Contextualisation of CDS

If the problems of a particular patient are noted down within a patient problem list, that contains the **established diagnoses** and the **unexplained findings**, this problem list can be used for the **contextualization of clinical decision support tools**. For each generic patient problem, there should be a predefined **problem solution strategy** in the form of at least one specific **problem-based clinical decision support tool** (CDST), which is a compilation of the clinical decision rules that need to be considered here.

2.6.4 Problem-based CDS Tools

In problem-based clinical decision support all decision rules that are needed to sufficiently manage a clinical problem are integrated into one clinical decision support tool.

A problem-based clinical decision support tool is supposed to request exactly all those patient data from the user or from other sources (e.g. the electronic health record), that need to be collected to demonstrate a good (e.g. guideline-conform) management of the patient problem.

The underlying assumption is, that all action statements of a particular guideline together sufficiently cover the underlying problem in its entirety. Obviously, this is not always the case. Guideline authors sometimes list gaps within the evidence of their suggested problem solutions in their clinical practice guidelines and then do not demonstrate an action statement and a decision rule. They also often leave uncovered areas where no actions (or alternative non-actions) are mentioned. These uncovered areas are sometimes not clarified to the user. If there is no guideline available for a particular patient problem or a suitable guideline is outdated or is not locally appropriate, alternative CDS tools that cover the clinical problem sufficiently can be set up by medical experts. A CDS tool is not necessarily bound to the existence of an appropriate clinical problem guideline.

2.6.5 CDS Tool Profile

As a particular problem-based clinical decision support tool is a compilation of applicable clinical decision rules, the rules that are to be integrated need to be collected and chosen. If a CDS tool is entirely and purely based on one clinical practice guideline (e.g. 2016 ESC clinical practice guideline on atrial fibrillation), all clinical decision rules of the particular guideline are integrated into the tool. If a CDS tool is not only based on one clinical practice guideline, e.g. local adaptations are amended to the tool, then sources, authors, and other criteria of this specific compilation need to be made transparent to the users.

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This is why a **clinical decision support tool profile** (CDST profile) needs be set up for each CDS tool by the main authors of the tool. A CDS tool profile should (at least) list the following items:

- Title of the CDS tool (e.g. atrial fibrillation)
- Short description of the tool
- **Original source** and its authors (e.g. ESC guideline, 2016)
- **Indications for use**, that is who and what the tool applies to: **patient group**, anatomical structure, **disease** or injury (e.g. atrial fibrillation)
- User groups
- **Digitized by** and reviewed by who (author of the CDS tool)

A **risk-benefit conclusion for a particular CDS tool** is based on the overall general risk analysis of the CDS system, referring to the risks and potential harms that are bound to demonstrating expert knowledge to healthcare professionals etc. On the other hand a **specific indication related risk-benefit conclusion**, the medical risk-benefit conclusion, relating to the particular action (recommendation) related medical risks, needs to be done for each clinical decision rule and needs to be demonstrated within the clinical decision rule profile.

A **clinical evaluation** of a particular tool should also be based on the overall general approach of evaluating the CDS system in its entirety, its standardized way of showing information, etc. A specific evaluation of what needs to be done medically will be examined on clinical decision tool and decision rule level.

2.7 Components of Clinical Decision Support Systems

2.7.1 Major Components

The **six major components** of an ideal problem-based clinical decision support system are:

- **Knowledge base**: decision rules with clinical input elements and clinical output recommendations (and compilations of these decision rules) and other non-CDS content
- **Inference engine**: processing decision rules of knowledge base
- **User interface**: communication mechanism to receive input and show output to user
- **Explanation component**: justifying recommendations to user for transparency
- **Patient database**: storing CDSS specific patient data
- **Admin User Interface:** CDS editor to allow experts to set up decision rules

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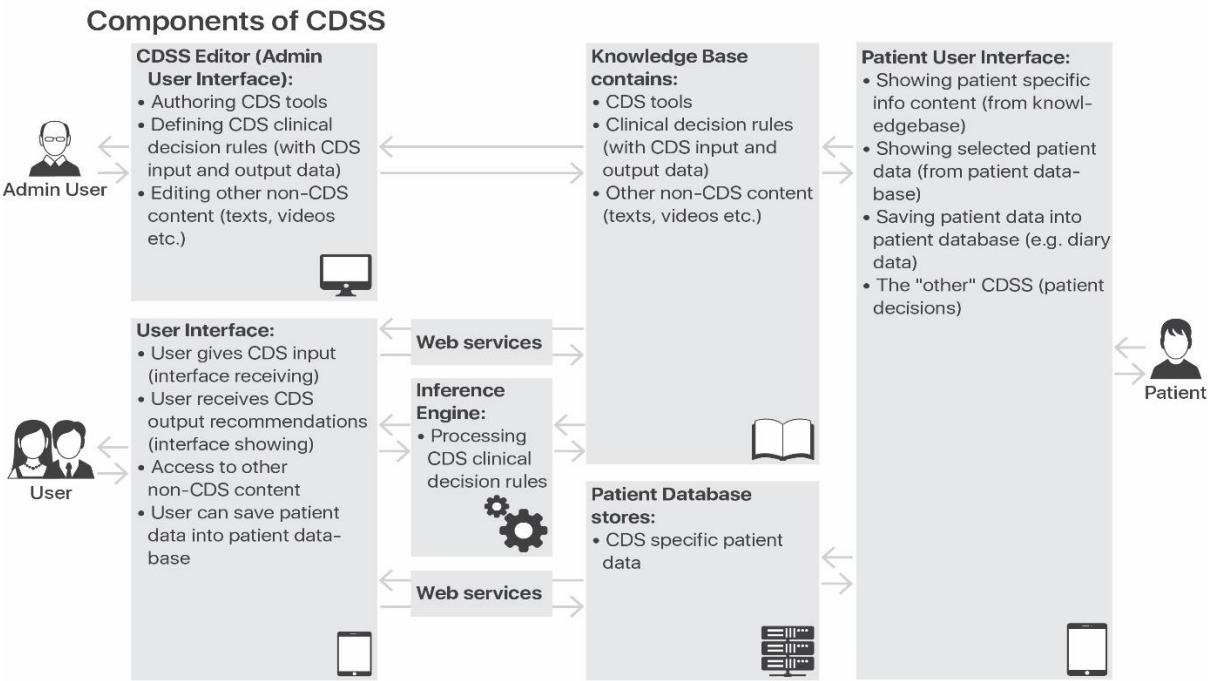

Fig 1: Components of a Clinical Decision Support System

The **knowledge base** contains a collection of decision rules, IF-THEN rules or condition action statements. The conditions of these decision rules are predefined structured **clinical input data** (terms, elements and modifiers). The actions are digitized **clinical output recommendations**. Based on specific patient data **input** of the user through the **user interface**, the **inference engine** combines the rules from the knowledge base with the patient data (input) and demonstrates one or more digitized recommendations (outputs). This automated reasoning is done in the background. The output is shown to the user through the **user interface** of the CDS system. To better understand the clinical output recommendations, its underlying reasoning and sources, a CDSS should contain an **explanation subsystem component** or justifier. Explanatory content can again be demonstrated through the user interface. Within a **patient database** the user is enabled to store CDS specific patient data. The **admin user interface** is a clinical decision support editor that allows clinical experts to transform analog medical knowledge, e.g. guideline statements into the production rule format of a CDSS.

2.7.2 Input

Clinical information is entered into the CDSS by the user selecting out of a **given set of input data** (predefined terms and modifiers) on the user interface. These input data serve as the IF-conditions of the clinical decision rules. Alternatively, input data can be automatically populated into a user interface out of other information systems (e.g. HER / EMR or patient diaries). Data that are automatically filled into CDS tools could be affirmed or negated (overruled) by the user (physician).

Input data (terms, conditions) primarily describe the clinical situation of the patient encompassing findings from the medical history, physical examination, laboratory results, and other pieces of clinical data [1]. Where necessary these findings can be described more detailed, e.g. by a vocabulary of severity, location modifiers or temporal modifiers. Calculators

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and medical scores can be used to objectify clinical conditions. These predefined input data fields are set up in a structured (digitized), machine readable or machine processable way, e.g. to allow the user to make yes/no choices or to type in numeric figures.

2.7.3 Output

Based on the input of the user (IF-conditions of decision rules) and on the automated reasoning in the background, one or more **digitized clinical recommendations** (outputs, THEN-actions) are demonstrated to the user by the CDS system. These recommendations need to be acknowledged or negated by the physician and further actions may thereby be approved (or disapproved). The output itself (e.g. a diagnosis) can again be input for a next step of reasoning and produce further next-level output (e.g. a treatment recommendation).

2.7.4 Computerized Physician Order Entry (CPOE), Order Sets

Computerized physician order entry (CPOE) is defined by the Healthcare Information and Management Systems Society (HIMSS) dictionary as an "order entry application specifically designed to assist clinical practitioners in creating and managing medical orders for patient services and medications" [3]. An **order set** is a pre-packaged group of related orders that apply to a specified diagnosis or a particular period of time and that a physician can easily place. Using order sets reduces both time spent entering orders and terminal usage.

Order sets and CPOEs of existing electronic health record systems can be used as input or output for clinical decision support systems.

2.7.5 Inference Engine

The **Inference engine** is the portion of the CDSS that combines the requested input data (IF-conditions) and other data according to the logical scheme (reasoning) of the decision rules for generating the output recommendations (THEN-actions) [1]. Whenever all required input data of a particular production rule are completely available to the CDSS, e.g. typed into the interface by the user, the rule can be executed and the output recommendation can be shown on the user interface.

2.7.6 User Interface of Clinical Decision Support

Patient data can be entered into the CDSS through the **user interface** and output is shown to the user through the **user interface**. The user interface is a communication mechanism of the CDSS to receive input and show output.

Generally clinical decision support and the results of clinical decision rules can be represented in different ways, e.g. just in the form of single **alerts** that pop up within an electronic health record when the conditions of the decision rule are met. A more complex way of demonstrating CDS is to integrate all decision rules that apply to one specific patient problem into one **user interface**, into a **problem-based clinical decision support tool**. Such problem-based CDSTs are collections of clinical decision rules that cover the solution of a particular clinical patient problem. They represent comprehensive problem solution strategies for a particular patient problem.

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In these problem-based CDS tools decision rules can be represented in one of two formats. If the decision rules are characterized by an explicit flow with a series of branching questions (IF-THEN logical statements), then they can be represented as **decision trees** or **flow charts**, often called **algorithms** or **procedure-based systems**. If the production rules cannot be arranged within explicit flow other ways for representation need to be used. In these **non-procedure-based systems** each unit of knowledge is a single IF-THEN-logical statement, and an inference engine, evaluating the available data and statements, chooses which statement to execute next. Both formats of problem-based clinical decision support have in common that they are based on these IF-THEN-logical statements.

2.7.7 Knowledge Base

The knowledge base of a CDSS is a library of clinical decision rules (IF-THEN rules or condition action statements) **and compilations of these decision rules** (e.g. problem-based CDS tools). The knowledge base primarily contains the production rules with all their clearly defined structured medical input elements and digitized output recommendations. The production rules can be aggregated within problem based clinical decision support tools in either procedure-based or non-procedure-based formats. Such an aggregation of patient situation specific decision rules is potentially also achievable by using meta rules for dynamically aggregating compiling the applicable decision rules and demonstrating them in on-the-spot-compilations through the user interface.

2.7.8 Explanation Component

For better interaction with users a CDSS should preferably contain an **explanation subsystem component** or justifier. The explanation component helps the user to better understand the clinical output recommendations, its underlying reasoning and sources. There should be structured explanatory notes for each production rule, ideally in the form of a **clinical decision rule profile** that summarizes clearly the underlying guideline statement or alternative decision-making processes e.g. a harm-benefit assessment. This explanatory content can be stored within the knowledge base and demonstrated through the user interface. The explanation component is a logic subsystem and part of other CDSS components (knowledge base and user interface). The explanation component improves transparency for users of the decision support system.

2.7.9 CDSS Editor

The **clinical decision support system editor** is an **admin user interface** that allows clinical experts to transform analog medical knowledge, e.g. guideline statements into decision rule formats of a CDSS. It may also allow local admin users to adapt given CDS tools to local settings. The CDSS editor also provides admin users to author explanatory contents (Non-CDS content e.g. texts, audios, videos) and store them within the knowledge base. The CDSS editor demands clinical experts to understand underlying CDS concepts like Boolean algebra, probability and decision theory. A CDSS editor however relieves clinicians from writing software code themselves or working directly with software developers.

2.7.10 Patient Database

The **patient database** enables users of a CDSS to store CDSS specific patient data, that might not be compatible to regular electronic health record systems, into a specific patient database.

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An agile architecture of the **patient database of a CDSS** also allows admin users to determine new CDSS input data, data sets and data derivatives in a quick way. As a CDSS patient database needs to fulfill special agile requirements, it seems reasonable to keep a CDSS patient database logically apart from conventional patient databases like electronic health records and other patient data archives.

3. Product Definition

3.1 Product Description

3.1.1 Overview

The ZAZA Pro system is a problem-based multidisciplinary clinical decision support system (CDSS) to be used for diagnosis, treatment planning and treatment through clinical decision support tools for specific clinical purposes and indications.

ZAZA Pro is a continuously updated medical device system, which integrates the customer's expertise in designing specific CDSTs into its product design and release processes resulting in up-to-date, verified, evaluated and validated clinical decision support (clinical knowledge) provided through ZAZA Pro CDS tools. The multidisciplinary character will increase with every version by adding further specific clinical decision support tools (CDSTs) to the initial CDSTs, such as AT01 "ESC 2016 Atrial Fibrillation Dashboard". See "Product History CDS Tools List" for all included CDSTs.

The included CDS Tool **AT01 'ESC 2016 Atrial Fibrillation Dashboard'** as first example can be described as follows:

- **Tool Description:** This CDS Tool is supporting the physician in decision making for diagnosis, treatment planning and treatment of patients suffering from atrial fibrillation using the rules defined in 'ESC 2016 Guidelines for the management of atrial fibrillation developed in collaboration with EACTS'.

All tool specific definitions, such as product description, intended use or indications for use are defined in the according **tool profile,** e.g. CDS Tool Profile Atrial Fibrillation ESC 2016 for AT01.

ZAZA Pro supports **admin users** (medical experts, medical associations) in transforming analog clinical knowledge (e.g. evidence-based clinical guideline texts) into **machine-readable input data** (patient data), **output data** (recommendations) and the underlying **clinical rules** (automated reasoning). It supports admin users to design these **problem-specific clinical decision support tools (CDSTs)** within the **ZAZA Pro CDSS editor** platform environment. ZAZA Pro provides functionality to ensure verification and validation including clinical evaluation of the clinical decision rules during creation of the customized CDS tools supported by the **ZAZA Pro CDSS editor**. ZAZA Pro allows the admin users to release these up-to-date specific CDS tools into the **ZAZA Pro point-of-care (POC) app**. ZAZA Pro thereby supports clinical experts (admin users) to better disseminate the CDS knowledge to healthcare professional users at their point-of-care. The CDS tools are available from an online library through a **ZAZA Pro POC app** or a **ZAZA Pro website**.

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ZAZA Pro supports healthcare professional **users** at the point-of-care (POC) in making their individual clinical patient decisions. **Input** and **output** are given by and to the user via a **graphical user interface**, mainly on mobile devices or within browser applications. Specific clinical knowledge about a particular clinical problem (e.g. a specific disease) is presented and grouped within **problem-specific clinical decision support tools.** The **ZAZA Pro POC app** allows users to use these CDS tools (CDSTs) and document results into a **ZAZA Pro patient database**.

3.1.2 Börm Bruckmeier's CDSS Business Model

Börm Bruckmeier's (BB) **value proposition** is to increase quality, safety, efficiency and transparency in clinical patient care through the ZAZA Pro clinical decision support system. BB licenses its ZAZA Pro CDSS through a CDS platform to healthcare professional users who work in hospitals (hospital chains, university hospitals, single hospitals) and in private practices. **Revenues** will be generated from **license fees** paid by **hospital and private practice customers** and by monetizing from data analyses. BB will **partner** with CDS tool-producing organizations (medical societies, key opinion leaders), consultants (implementation management, distributors), evaluators (evaluation of cost and quality) and other non-competing providers (EHRs, CME).

3.1.3 ZAZA Pro Description

ZAZA Pro is the medical device covering server and client software system components and specific problem-based clinical decision support tools (CDSTs).

The server software is mainly a **CMS editor** (admin user frontend) for the creation of CDS tools and other non-CDS content (e.g. texts, videos), a **knowledgebase** to store CDSTs and other content and a **patient database** on a different server. The client software component is mainly the mobile framework **point-of-care (POC) app** for the demonstration of **clinical decision support tools (CDSTs)** and other medical content.

ZAZA Pro is a **CDS system (CDSS)**, consisting of:

- **ZAZA Pro platform** (server components):
  - CDSS editor: to produce CDS tools, non-CDS tools and other knowledge content
  - Knowledge base: to store and manage CDS tools, non-CDS tools and other knowledge content
- **ZAZA Pro point-of-care (POC) apps** (client components):
  - Mobile POC app: to use the CDS tools and other content at the point-of-care
  - Website version of POC app
  - Intranet server version of POC app
- **Patient Database** (server component) and
- Multiple **specific ZAZA Pro clinical decision support tools** (**CDSTs**)

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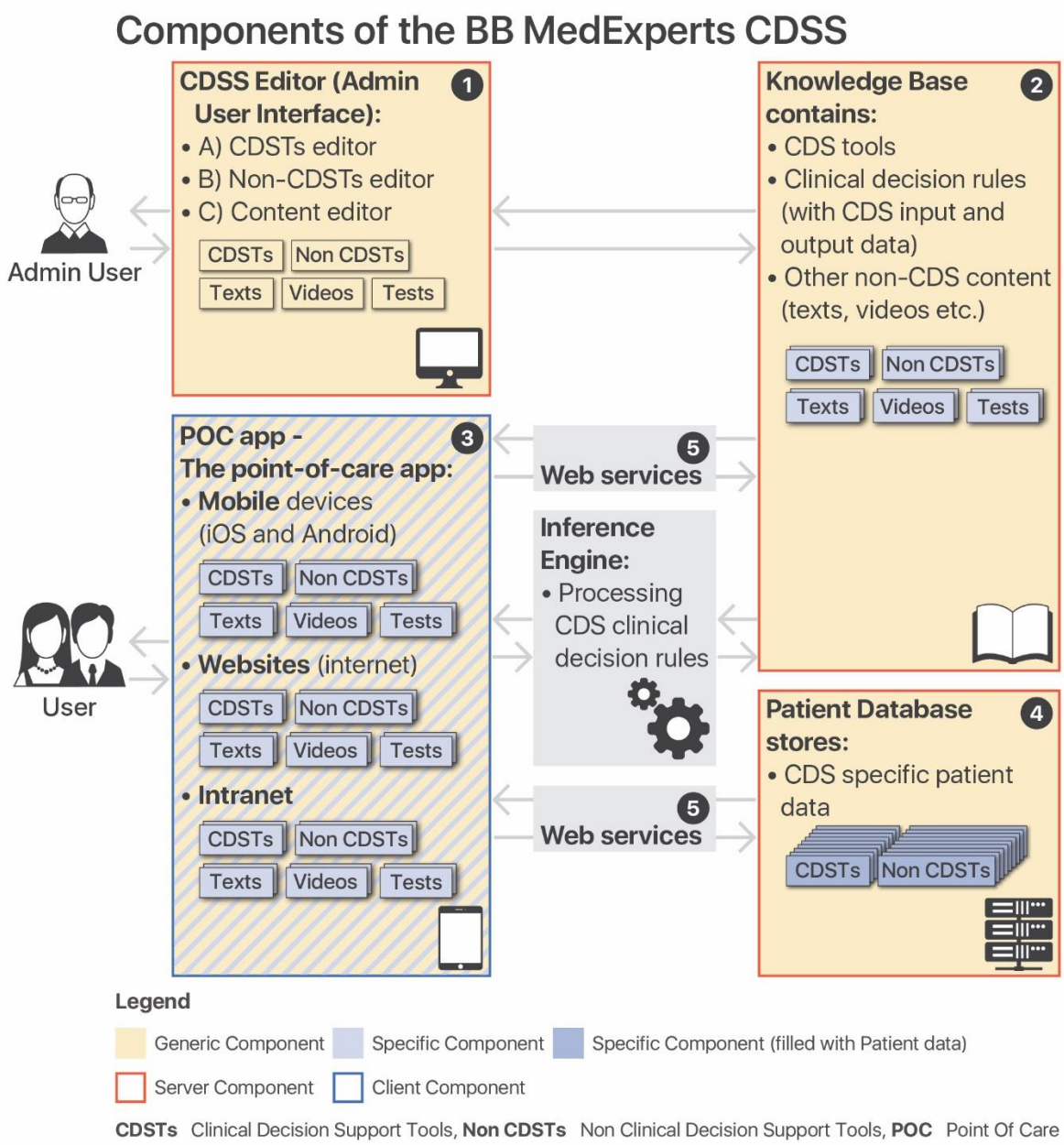

Fig 2: Components of the ZAZA Pro CDSS

3.1.4 Technical Components, Functions

The technical components explanations of the ZAZA Pro CDSS:

1. **CDSS editor**
   A) CDSTs editor: The CDSS editor allows admin users (e.g. clinical experts, BB editors) to create clinical decision support tools (CDSTs: clinical process protocols, dashboards), which means to define the needed structured clinical input data (input), the clinical output recommendations for actions, and the automated reasoning, the underlying rules that connect the input data with the applicable output data. The CDSS editor supports admin users to set up a separate decision rule profile for each clinical decision rule including a specific clinical evaluation. The decision rules can then be compiled into one problem-based clinical decision support tool. The CDSS editor supports admin users to also set up a separate clinical decision tool profile for each CDST. The CDSS

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editor then allows review and approval of each clinical decision rule and its compilation into a CDST by a different admin user. Only after its verification and validation, a CDST is qualified to be released into the ZAZA Pro knowledgebase.

B) Non-CDSTs editor: Other interactive tools which do not contain automated reasoning (Non-CDSTs: calculators, scores, checklists) and which are therefore not medical device relevant, can also be set up within this CDSS editor. These tools are marked as **non-CDSTs**.

C) Content editor: The CDSS editor also allows admin users (e.g. clinical experts, BB editors) to create and place other content into ZAZA Pro, that is not a tool (CDST, Non-CDST), e.g. texts, audios, videos and tests.

2. **Knowledgebase** - The knowledgebase server stores all tools (CDSTs and Non-CDSTs) and other content, which were created by admin users within the CDSS editor. The knowledgebase serves as a repository for users to choose and download applicable CDS tools and other content onto their POC devices into the POC app.
3. **POC app** - The point-of-care app (POC app) provides a user interface for healthcare professionals to consume/download the content and tools (from the knowledgebase server), to then interact with the content and tools (locally) and to view, edit and generate patient data (from/on the patient database server). The POC app is developed for **mobile devices** (iOS and Android), **websites** (internet) and **intranet** versions.
4. **Patient database** – The patient database server provides a data repository for storing and retrieving patient data (e.g. needed for CDSTs and POC app).
5. **Web services** – Web service components serve to support interoperable machine to machine interaction for exchanging data between the CDSS editor, the knowledgebase server, the POC app and the patient database over a network.
6. **Multiple specific ZAZA Pro clinical decision support tools** (**CDSTs**): many clinical decision support tools (CDS tools) running on mobile devices and desktops at the point-of-care (e.g. AT01 Atrial Fibrillation Dashboard). The CDS tools are limited to the two types: clinical process protocols (procedure-based CDSTs) and dashboards (non-procedure-based CDSTs).
7. **Other components** – Other components like network security and database encryption, regular backups and restoration procedures, user feedback systems and IT infrastructure are mandatory for the system to be up and running or are additional services of the system.

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3.1.5 ZAZA Pro CDS Tools

A **ZAZA Pro CDS tool** is generated by a **medical expert** (admin user) within the **ZAZA Pro CDSS editor**. The required clinical knowledge (clinical decision rules and medical reference content) is provided by medical associations, international medical experts or individually by single customers and is incorporated either by the manufacturer or the customer. Each ZAZA Pro CDS tool is a collection of problem-based clinical decision rules. For each decision rule, the admin user is required to set up a separate **decision rule profile** including a specific **clinical evaluation**, which is used together with all specific clinical evaluations for each CDST for a comprehensive clinical evaluation for the ZAZA Pro system as a whole. For each clinical decision tool, the admin user is required to set up a separate **clinical decision tool profile**. Each decision rule and the compilation of all clinical decision rules into a ZAZA Pro CDS tool need to be **reviewed** and **approved** by at least one additional (second) admin user. Only after **verification** and **validation** of the ZAZA Pro CDS tool, the tool can then be released into the **ZAZA Pro knowledgebase**. The CDS tool can then be downloaded by users (e.g. physicians) from the knowledgebase into the **ZAZA Pro POC app**. It can then be used to support individual patient treatment decisions of the physician at the POC. CDS tool results can be saved into a **ZAZA Pro patient database**.

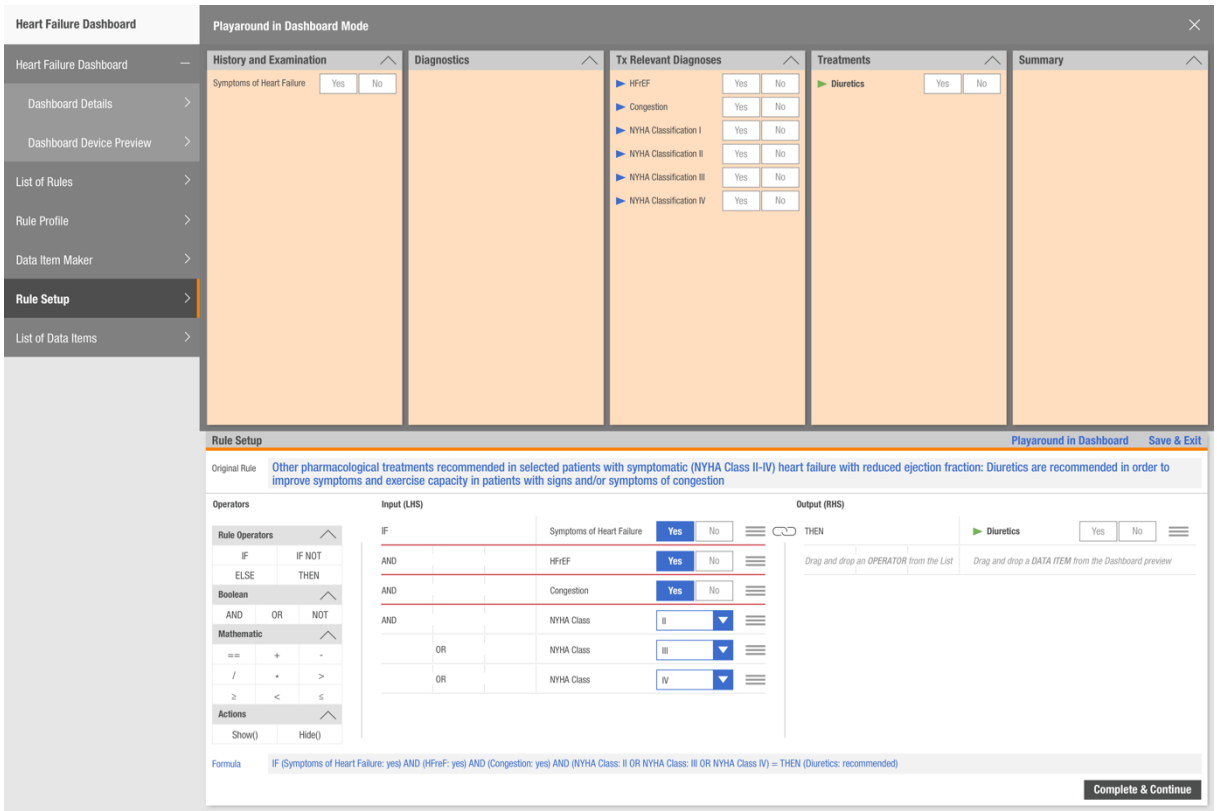

Fig 3: Representation of a clinical decision rule

A **ZAZA Pro CDST** is a transferrable collection of problem-based clinical decision rules. Within the environment of the ZAZA Pro POC app, the ZAZA Pro CDS tool becomes an executable program. A ZAZA Pro CDS tool can be regarded as a problem-specific CDSS knowledgebase, a CDSS user interface and a CDSS inference engine at the same time. A list which contains all ZAZA Pro CDSTs, including reference to the tool definition is stored within the "**Product History CDS Tool List**".

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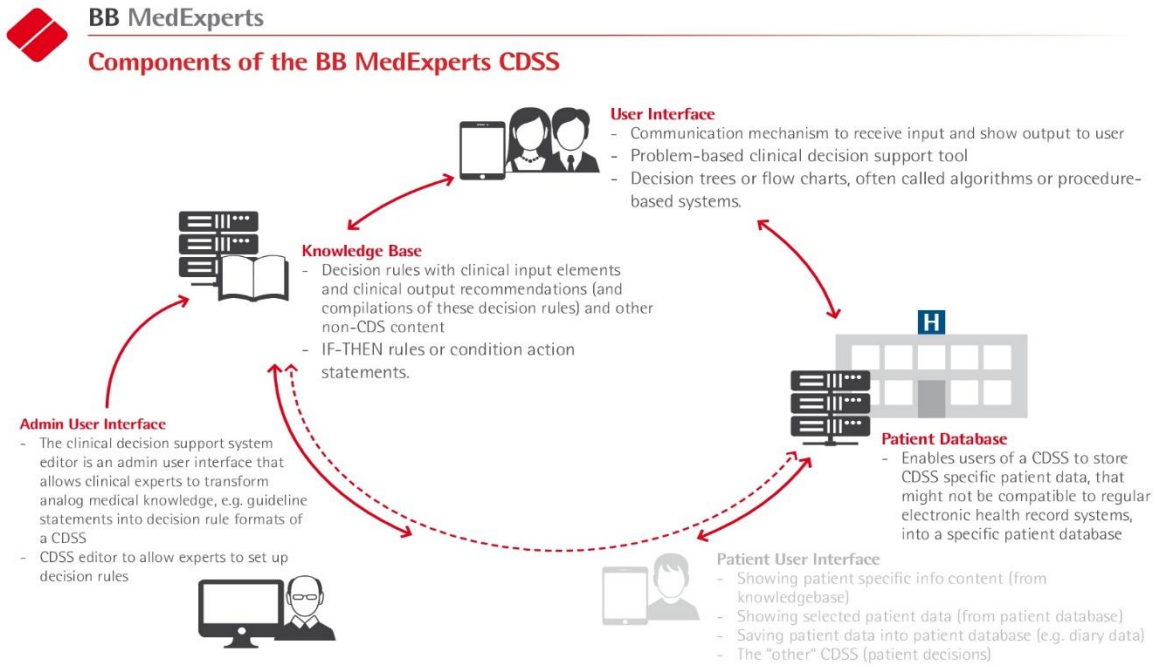

Fig 4: Components of the ZAZA Pro CDSS

3.1.6 Formats of ZAZA Pro CDS Tools

Within the ZAZA Pro problem-based multidisciplinary clinical decision support system decision rules are also demonstrated in two different formats: a **procedure-based way**, this is called a **clinical process protocol** (CPP, algorithm, procedure), and a **non-procedure-based way**, which is called a **clinical dashboard**.

A **clinical process protocol** (CPP) is a procedure-based system, in which **all decision rules** of a particular patient problem can be arranged in **one explicit flow** and can be shown in a decision tree or a flow chart. Apart from the demonstration of the flow (algorithm) and from referring to the clinical decision rule profiles via links, clinical process protocols allow for automated documentation: everything that has been chosen and done by the physician can be saved within the ZAZA Pro patient database. The terms and structured data that are saved can be predetermined.

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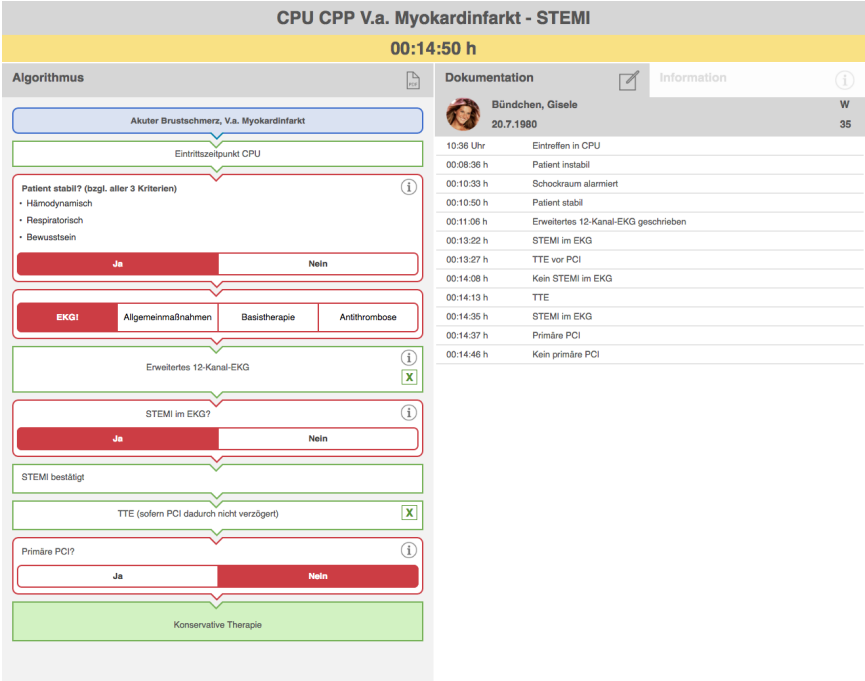
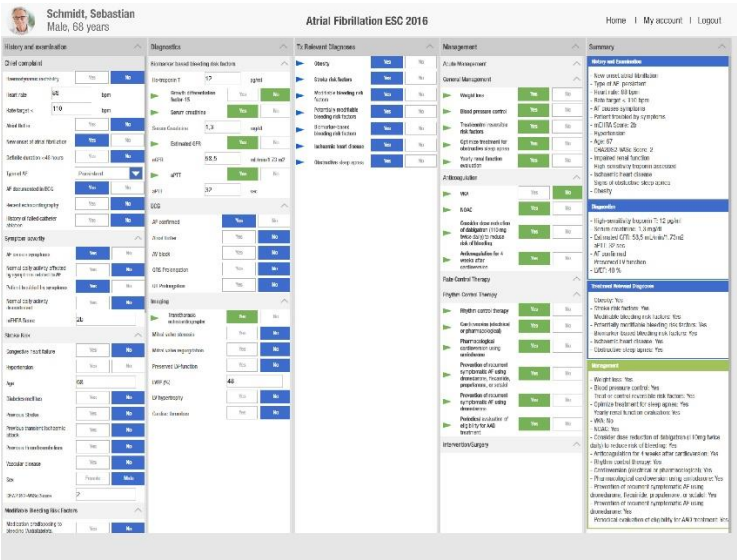


Fig 5: Sample of a ZAZA Pro Clinical Process Protocol

Within ZAZA Pro a clinical dashboard is a non-procedure-based CDS tool in which all decision rules that are needed for the solution of a particular clinical problem are spread over a matrix that is divided into logical sections. Such a matrix can be grouped e.g. into the four sections history/physical exam, diagnostics, diagnoses and treatments. Each clinical decision rule needs to be placed into the matrix by grouping its IF-condition (medical input data) into one of the sections and also its THEN-recommendation (output data) into one of the sections. If-condition and THEN-recommendation of a particular decision rule are usually placed into neighboring sections of a dashboard matrix, but can also be grouped into the same section (e.g. both within the diagnostics section) or into non-neighboring sections (e.g. physical exam and treatment section). **A clinical dashboard is a matrix for a particular patient problem that incorporates all treatment relevant clinical decision rules by demonstrating all required structured input data (medical elements) and all determined recommendation output data**.



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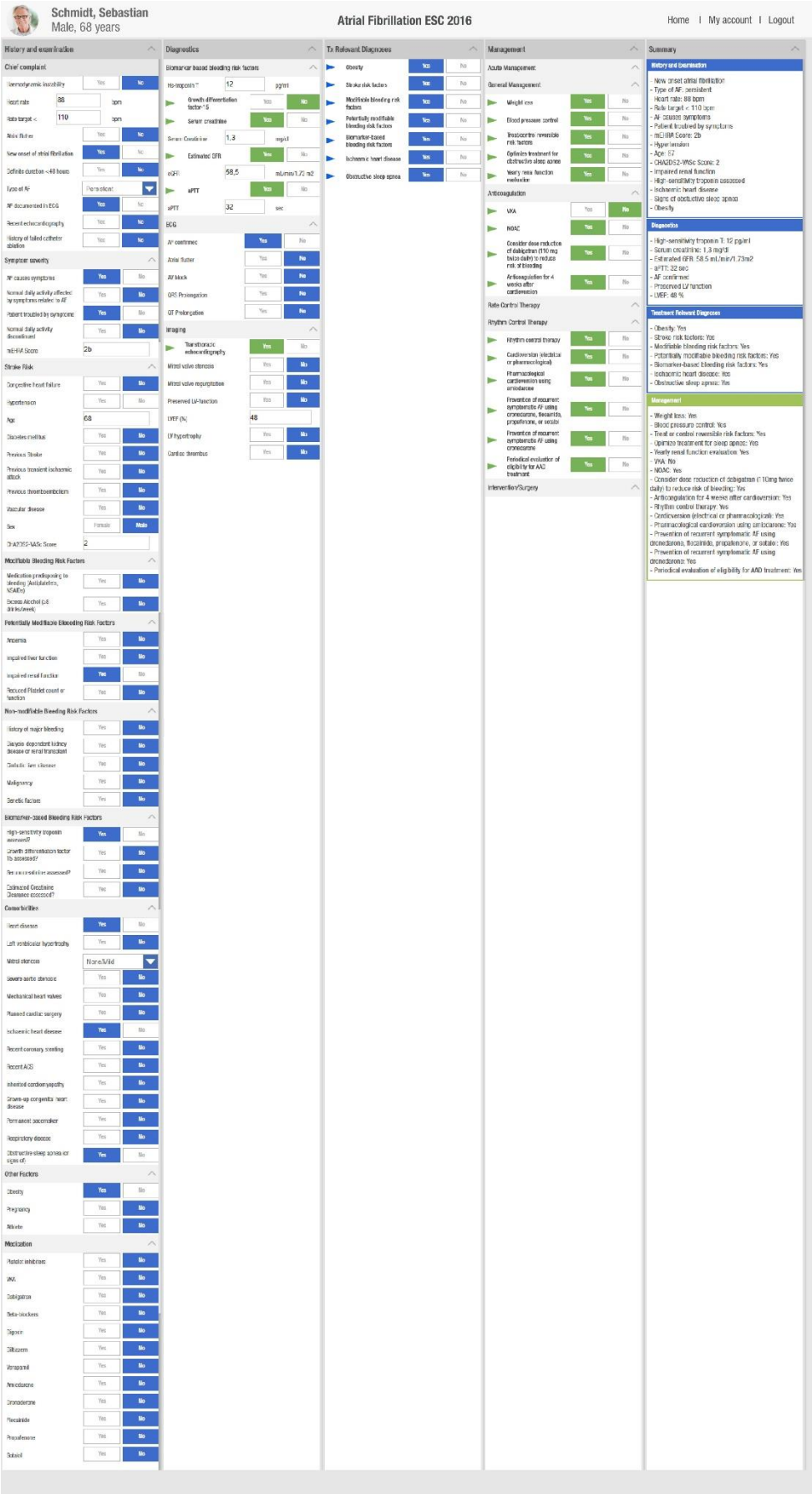

Fig 6: Sample of a ZAZA Pro Dashboard

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When using a ZAZA Pro dashboard, a clinical decision rule is immediately executed and the THEN-recommendation is shown to the user, after all input data of a particular decision rule have been fully specified. Not all decision rules and its IF-input data are shown to the user right away or at the same time, but only those rules and its IF-conditions are shown that need to be processed at that time. E.g. a decision rule that demonstrates a particular treatment will not be shown before all relevant diagnostic data are figured out. THEN-recommendations are only demonstrated, when all needed IF-input data of that particular rule were given into the CDSS. Alternative recommendations, that were, based on the input data, not chosen by the system, are by default not shown at all.

3.1.7 Production of ZAZA Pro CDS Tools

When **transforming a problem-specific guideline** into a particular ZAZA Pro **problem-based CDS tool** on the ZAZA Pro CDSS editor, an admin user (e.g. a guideline author or a BB editor) transcribes the given analog clinical knowledge into the needed discrete digital input and output data elements. First, all metadata of the guideline are copied into the ZAZA Pro CMSS editor: guideline producing organization (author), year of publication, original source, etc. Then, all analog executable guideline action statements are extracted out of the clinical practice guideline text and transferred into a list. Then all needed IF-conditions and all THEN-action recommendations of one statement are extracted, structured and digitized by determining them in a digital format. These digital forms of a decision rule, its IF-condition inputs and THEN-action outputs then need to be positioned into the right sections of the CDS dashboard or the algorithm-style clinical process protocol. The digital input and output data of a decision rule should then be connected by using logical operators like conjunction, disjunction, negation, and also other mathematical concepts of set theory, Boolean algebra, numeric algebra, probability and decision theory. For each clinical decision rule of a problem-based CDS tool, a **complete decision rule profile** needs to be filled out. And for the problem-based CDS tool itself, a **complete CDS tool profile** needs to be filled out.

3.2 Medical Purpose, Intended Use

3.2.1 Intended Use

The **purpose of this ZAZA Pro CDS system** is to support physicians in managing clinical knowledge and in making clinical decisions, including diagnosis and treatment decisions at the point of care during clinical routine and under consideration of actual patient data through Clinical Decision Support Tools (CDST).

The included CDS Tool **AT01 'ESC 2016 Atrial Fibrillation Dashboard'** as first example is intended as follows:

- **Intended use: Support the physician in decision making for diagnosis, treatment planning and treatment of patients suffering from atrial fibrillation** using the rules defined in 'ESC 2016 Guidelines for the management of atrial fibrillation developed in collaboration with EACTS'.

All tool specific definitions, such as product description, intended use or indications for use are defined in the according **tool profile,** e.g. CDS Tool Profile Atrial Fibrillation ESC 2016 for AT01. All CDS tools and its profiles are listed within the "**Product History CDS Tool List**".

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The ZAZA Pro CDS System allows the user to:

- manage (including standardization and support of process validation) clinically relevant knowledge (experience) within a knowledgebase
- create customized clinical decision support tools including that knowledge within a CDSS editor
- easily retrieve this knowledge at the point of care (POC) via mobile device
- document the performed clinical decisions and processes into a patient database
- collect patient data in a patient database

Supporting the physician is understood as providing objectified knowledge to the physician in specific patient situations to:

- **aid the physician's clinical decisions and processes** (including diagnostic and therapeutic decisions) at the point of care
- determine medical **conditions** of the patient through standardized condition criteria, predetermined IF-conditions of clinical decision rules
- make suggested clinical **decisions** following clinical decision rules
- perform and document specific clinical **activities** following defined standard operating procedures
- demonstrate the whole **process** (which is the chronological, logical sequence of these conditions, activities and decisions) in a standardized and interactive form, with clinical process protocols and dashboards

Standardization and management includes codification and optimization of the clinical relevant knowledge using an easily transferable, structured, machine-readable form, which is used by standardized interactive CDS tools providing easy to understand outputs (display) for the physicians.

3.2.2 Indications for Use

ZAZA Pro is considered to be a **multidisciplinary medical device**. And there is no limitation of particular indications for use of the system with respect to particular patient groups, anatomical structures, injuries, or particular medical fields.
The use of the system is indicated for any clinical organization providing health care or clinical treatment using validated clinical processes and validated clinical decision-making rules.

Furthermore, the ZAZA Pro includes and provides **specific CDS tools** for particular medical fields and **specific medical indications**.

The first included CDS Tool **AT01 'ESC 2016 Atrial Fibrillation Dashboard'** as first example is indicated as follows:

- **Indication:** Atrial Fibrillation, Atrial Flutter
- **Indicated Anatomical Structure/ Body Part:** Heart, Cardiovascular System
- **Indicated Patient Population/Group:** Adults (including pregnant women)

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All tool specific definitions, such as product description, intended use or indications for use are defined in the according **tool profile,** e.g. CDS Tool Profile Atrial Fibrillation ESC 2016 for AT01. All CDS tools and its profiles are listed within the "**Product History CDS Tool List**".

3.2.3 Use Context, User Groups and User Needs

The user of the System separates in **two main user groups**:

- admin users (authors, medical experts, knowledge) providing the knowledge
- users (knowledge receivers) receiving the knowledge

The users (knowledge receivers) are again split in two groups:

- healthcare professionals (professional app users)
- patients (patient app users)

Both receiving defined information in accordance to their needs as defined by the admin user (author, medical expert, knowledge).

Potential CDS Tool specific user groups or refinements are defined in a CDS tool **specific CDS tool profile**.

The included CDS Tool **AT01 'ESC 2016 Atrial Fibrillation Dashboard'** as first example is defined as follows:

- **Intended User Group:** Medical Doctors
- **Intended Use Context/Scenario/Limitations:** Use by medical professionals in a hospital or outpatient setting

All tool specific definitions, such as product description, intended use or indications for use are defined in the according **tool profile,** e.g. CDS Tool Profile Atrial Fibrillation ESC 2016 for AT01. The list of tools with tool profile is stored within the "**Product History CDS Tool List"**.

These user groups are systematic and not necessarily different persons, especially with respect to admin user and professional app user. In clinical organizations, a single physician or a group of physicians may be part of both user groups at the same time.

More important is the understanding, that the clinical relevant knowledge is created and provided by the admin users (authors) of ZAZA Pro themselves within the ZAZA Pro CDSS editor. The admin users (authors) can be any organization or group of physicians (e.g. national health services, medical associations, hospital physicians) or even single physicians. This is to preserve the therapeutic freedom of physicians. As explained in chapter exclusions, ZAZA Pro itself is not intended to create or adjust the authors clinical knowledge.

Besides the two main user groups, we also consider the patients (not being user) and the technical admin as user groups. Please see the table below.

User Groups	User Needs	Usage and Use Context
Admin User (author, knowledge provider), e.g. National Health Services and institutions (governmental	Ensure and improve quality clinical outcome.	**Admin users** provide and manage the knowledge (clinical decision rules, CDS

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and non-governmental), medical Associations, hospitals (hospital groups, university hospitals), other groups of physicians, single physicians		tools) within the CDSS editor using a PC client.
Users (professional POC app users): healthcare professionals (knowledge receivers)	Provide and improve the clinical treatment supported by ZAZA Pro	**Professional App User**, mainly physicians, request and receive the knowledge (clinical CDS tools) at the point of use, using a mobile platform and the ZAZA Pro POC app including customized clinical decision support tools (CDSTs). Beforehand the App User is installing the App on the mobile device.
Patient App User (Patient, Knowledge Receiver)	Be informed and contribute in reaching the desired health condition.	**Patient App User** request and receive the knowledge (patient information) and provide particular own patient data at the point of use, using a mobile platform and the ZAZA Pro Patient App including customized tools. Beforehand the patient app user is installing the app on the mobile device.
Patient (not being user)	Reach the desired health condition.	The **patient** receives clinical treatment as part of clinical processes, which are supported by ZAZA Pro. There is no direct interaction between ZAZA Pro and the patient.
Technical Admin	Ensure a safe and running system.	The technical admin is installing and operating ZAZA Pro technically using according clients to control the server components for any technical reason.

3.2.4 Exclusions

The system is not intended to create or adjust any clinical relevant knowledge. The clinical relevant knowledge must be solely created and provided by the physicians themselves to preserve the therapeutic freedom of physicians. That includes required information required for complete product verification, product validation and clinical evaluation, which is used during the creation of ZAZA Pro CDS tools to ensure a compliant technical documentation.

3.3 Principle of Action and Function

3.3.1 Principle of Action (clinical)

The generic clinical principle of action is supporting the clinical routine by providing relevant and high quality clinical information at the point of use.

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Potential CDST specific principle of action definitions or refinements are defined in a CDST specific **ZAZA Pro CDS tool profile** for every CDS tool. The list of tools is stored within the "Product History CDS Tool List".

This generic principle of action is realized as follows: The admin users, referred as partner or customers in the graphic, are introducing the clinical knowledge into the knowledgebase, referred to as BB Content Server in the graphic.

The professional POC app users, referred to as physicians (customers) in the graphic receive the knowledge and perform their clinical activities, which includes creation of patient data to be stored in the ZAZA Pro patient database or connected hospital systems.
The patient app user, referred to as patients in the graphic, also receive patient specific information and may produce their own patient data, collected in the ZAZA Pro patient database.

Specific clinical principle of action of each clinical decision rule is defined in the CDS rule **specific CDS rule profile**. All specific clinical principles of action that are used within a problem-based CDS tool are included in a CDS tool list. The list of tools with CDS rule profile is stored within the "Product History CDS Tool List".

3.3.2 Principle of Function (technical)

The generic technical principle of function separates in server and client systems. The server based knowledgebase and the CDS editor system, as well as the client system realized as POC app including standard and customized software tools.

ZAZA Pro provides the following actions and functions for the user to perform the following:

- Server based knowledgebase and the CDS editor system
  - Create and edit clinical decision processes (knowledge) in an online system, e.g.
    - own text/document
    - own CDSTs based on the provided CDST types
  - Push these CDSTs, non-CDSTs and other content into the POC application without human intervention
  - Save patients/physician data onto a server
  - Save results/paths of CDS tools onto a server
  - Recreate and present this data and CDS tools into an application without human intervention
  - Do this in a secure and encrypted way
- POC App and CDS tools
  - Download and Update the mobile app including the customized tools e.g. CDSTs
  - Use the clinical processes provided by the system

Specific CDS rule specific principle of function definitions or refinements are defined in the CDS rule **specific CDS rule profile** for every CDS rule. The list of tools which includes tool definition is stored within the "Product History CDS Tool List".

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3.3.3 Key Performance Characteristics

The following key performance characteristics are defined as:

- System (= information) of 365/7/24
- Fast response time for fluent interaction
- Multiplatform support browser, iOS, Android
- Content verification and validation support

Potential CDS tool or rule specific clinical performance characteristic (CPC) definitions or refinements are defined in a CDS tool specific CDS tool profile for every CDS tool. The list of tools includes tool definition is stored within the “Product History CDS Tool List”.

3.3.4 General Clinical Benefits

General benefits of using ZAZA Pro:

- Centralization and Standardization of knowledge
- Availability of knowledge at the point of care (point of use)
- Acceleration of implementation of the knowledge (higher and quicker penetration)
- Quality management tool
- Verification and validation of processes management
- Teaching tool

All major benefits lead to the following clinical benefits:

- Reduction of overall duration of treatment
- Increased quality (reliability and effectiveness) of treatment

Most important clinical evidence for clinical benefit (clinical studies), databases and registers:

- Pubmed/Medline
- Cochrane Central Register of Controlled Trials
- Database of International Guideline Network
- Adverse event database MAUDE
- Clinical study register CENTRAL
- ClinicalTrials.gov
- UpToDate®
- WHO Global Health Library
- Devices@FDA catalog of cleared and approved medical devices

3.3.5 Particular Clinical Benefits of Variants

N/A

3.3.6 Risk Considerations

The intended risk management concept is aligned with the intended use of ZAZA Pro, especially considering the particular location of ZAZA Pro within the overall clinical use context as shown by the following figure.

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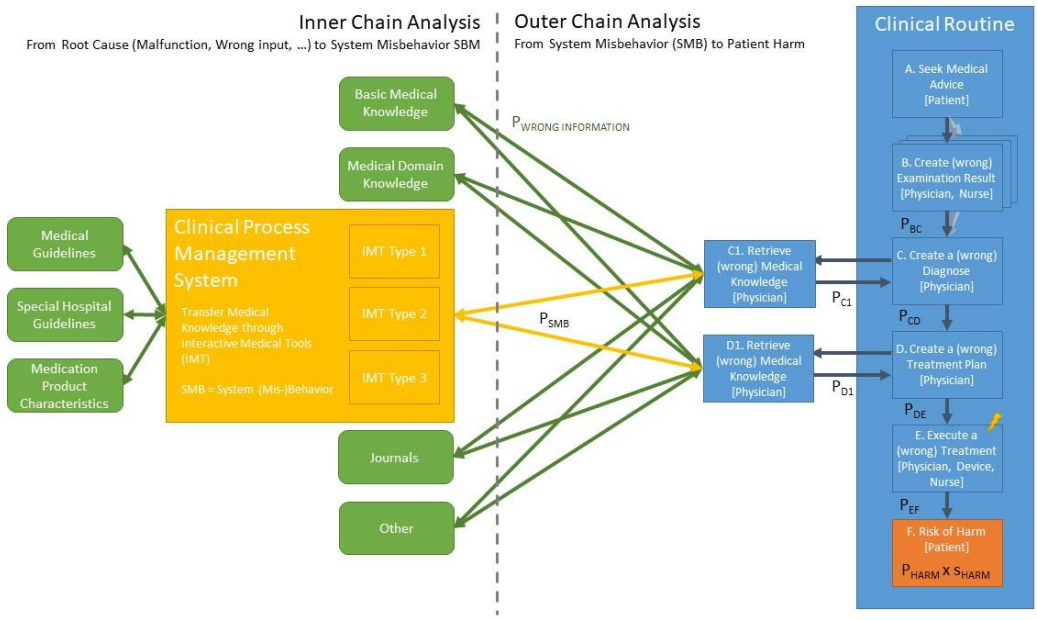

Fig 7: Risk management concept of ZAZA Pro

The figure above helps to understand the following key risk considerations:

1. The product, as a standalone software, is far away from actual harm.
    a. The product is especially not influencing any devices used for treatment, but only the information/knowledge retrieval done by the physician.
    b. The product is providing (partly) patient specific recommendations in real clinical routine cases, and is accordingly considered a medical device.
2. The product enables an alternative approach for providing information to the physician at the point of use, as an alternative to traditional ways, such as reading a guideline document in the hospitals bibliotheca.

The overall clinical use context shows that the risk of patient harm, especially the probability of occurrence of a patient harm is formed from a potentially very long chain of events, which consists of basically three sections.

- Events within clinical routine (blue area, A-F)
- Process of information retrieval (C1 and D1, blue boxes e.g. supported by ZAZA Pro, orange)
- Sources and quality of Information (green boxes and arrows), as part of the retrieval of information (C1 and D1)

3.3.6.1 Clinical Routine

The most important chain of events, which is mainly forming the risk of harm, is the clinical routine. The clinical routine is described in very generic way starting from A. patient is seeking medical advice ending with F. risk of patient harm. Each transition of this chain comes with a potential probability of occurrence of a wrong result. A multiplication of all these probabilities is the final probability of harm, P_{HARM}.

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Most of the transitions, especially the transitions after creation of a treatment plan, are out of influence of ZAZA Pro. ZAZA Pro is involved in the process of retrieving information, especially in C. and D., when the physician is creating a diagnosis or a treatment plan.

3.3.6.2 Process of information retrieval and Quality of information

The retrieval, respectively the quality of the information is again independent from ZAZA Pro in general. The physicians are using different sources with varying quality, as follows:

- Basic and specific knowledge the physician provides, simply by his education being provided as basic knowledge (in the head of the physician)
- Guidelines from various sources, such as organizations, health care provider, pharmaceutical companies or the hospital, provided in various ways, books, leaflets, processes
- Any other information from journals, or others

All this information comes with an incorporated risk, respectively probability of being wrong information, $P_{\text{WRONG INFORMATION}}$.

The final probability of occurrence of the "retrieval of wrong information" (e.g. P_{C1}) results from both, the probability of wrong information and the probability, that the process of information retrieval is erroneous, such as:

- remembering
- reading guidelines or books
- asking colleagues on guidelines
- reading or not reading journals

3.3.6.3 Risk management Concept

To ensure an effective risk management, we separated the risk management in two focus areas.

1. Inner Chain Analysis (System Misbehavior Analysis)
2. Outer Chain Analysis (Patient Risk Analysis)

Both together are required to estimate the risk of harm.

3.3.6.4 Inner Chain Analysis (System Misbehavior Analysis)

The Inner Chain Analysis focuses on the risks for system misbehaviour, such as a wrong recommendation. The followings are considered:

- Internal malfunction and errors
- Data inputs, especially the contents
- User inputs, during usage

By performing a FMEA type of analysis, we define according risk control measures to avoid the system misbehavior, such as wrong recommendations.

Main means of quality assurance and risk control:

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- Content validation:
  Incorporated mechanism to avoid usage of non-validated content (guidelines)
- Content verification:
  Incorporated verification mechanisms, to verify correct integration of contents
- Usability Considerations at the inputs:
  Incorporated usability functions (for the user input masks)
- SW development process (not a particular RCM):
  All SW development processes, such as software verification, as general quality assuring measures

All above is focused to avoid a system misbehavior as one of many contributing to a potential harm at the end of a subsequent outer chain of events.

3.3.6.5 Outer Chain Analysis (Patient Risk Analysis)

The analysis of the outer Chain starts from system behavior as a “side chain” of the clinical routine and ends with a harm, or the risk of a harm, at the end of the clinical routine.

The outer chain analysis focusses on identifying the applicable outer chains and define possible risk control measures, once the system misbehavior already happened. The outer chain analysis considers:

- the overall clinical use context (see figure above)
- all system (mis-)behaviors and the resulting outer chains
- chains of clinical routine (without medical product) as a comparison

by performing a combined analysis including FMEA and FTA type of analysis.

As a matter of fact, the means for control of the risk, with a system misbehavior on place are limited, due to the nature of the product. The following means of quality assurance and risk control for the outer chain may apply.

- Usability Considerations at the outputs:
  Incorporated usability functions (for the user output screens)
- Warnings and Information:
  Incorporated functions, making user aware of residual risks

Besides the risk control measures related to our product, the outer risk chain analysis also considers external risk reduction measures from clinical routine such as:

- Clinical routine for multiple examinations
- Clinical routine for confirmation by alternative diagnosis
- Clinical routine for patient information
- Clinical routine for patient monitoring during treatment
- Any safety routines incorporated in clinical routine

A main aspect of outer chain risk analysis is to evaluate the level of influence compared to potential other errors, that may result in harm.

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3.3.6.6 Summary of Risk Management Concept

Our risk management

- is not intended nor able to identify risks, related to any particular content.
  The customer is responsible for the content to be provided by ZAZA Pro.
  → Nevertheless, we provide verification and validation functions to support the customer.
- is not intended nor able to evaluate and analyses the particular hazardous situations or harms of particular treatments, which are executed after diagnosis or treatment plan was created supported by ZAZA Pro, in detail.
  → Nevertheless, we use generic estimates and the worst-case approaches to provide a complete risk chain.
- is intended to identify all risk control measures to avoid system misbehavior. That includes the analysis of the following functions and actions:
  - Risks resulting from server-side CMS functions and actions, such as:
    - Data handling
    - Data interaction
    - Data management (import, maintenance, adjustment) procedures
    - Tools creation
  - Risks resulting from client-side mobile app, such as:
    - App malfunctions
    - App usability
  - Risk resulting from standard and CDST types:
    - Usability or malfunction of clinical process protocols
    - Usability or malfunction of clinical dashboard

3.3.7 Cyber Security Strategy

The nature of the product characterized by server and client architecture, by combining web services and local client software and by handling clinical and private patient information, requires an up to date cyber security strategy to ensure IT security.

We follow the state of the art guidances e.g. cyber security guidance (FDA), Medizinproduktebetreiberverordnung, to ensure the handled data are:

- Integrity
- Availability
- Confidentiality
- Accountability
- Authenticity

According requirements addressing the IT security are incorporated in the requirements specification as an integral part. External attacks and intended misuse is considered during risk management.

3.4 Technical Data

3.4.1 System Architecture and Main Components

The ZAZA pro system is technically specified as follows and refined in the chapters below:

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1. IT System
2. Server-Client Architecture
    a. Server Components
    b. Client Components (e.g. mobile POC app and CDSTs)
    c. Databases
3. Technical Characteristics
    a. Safe and secure (e.g. by SSL)
    b. Web technologies (e.g. PHP, HTML, ...)
    c. Mobile technologies (e.g. iOS, Android, HTML5, ...)

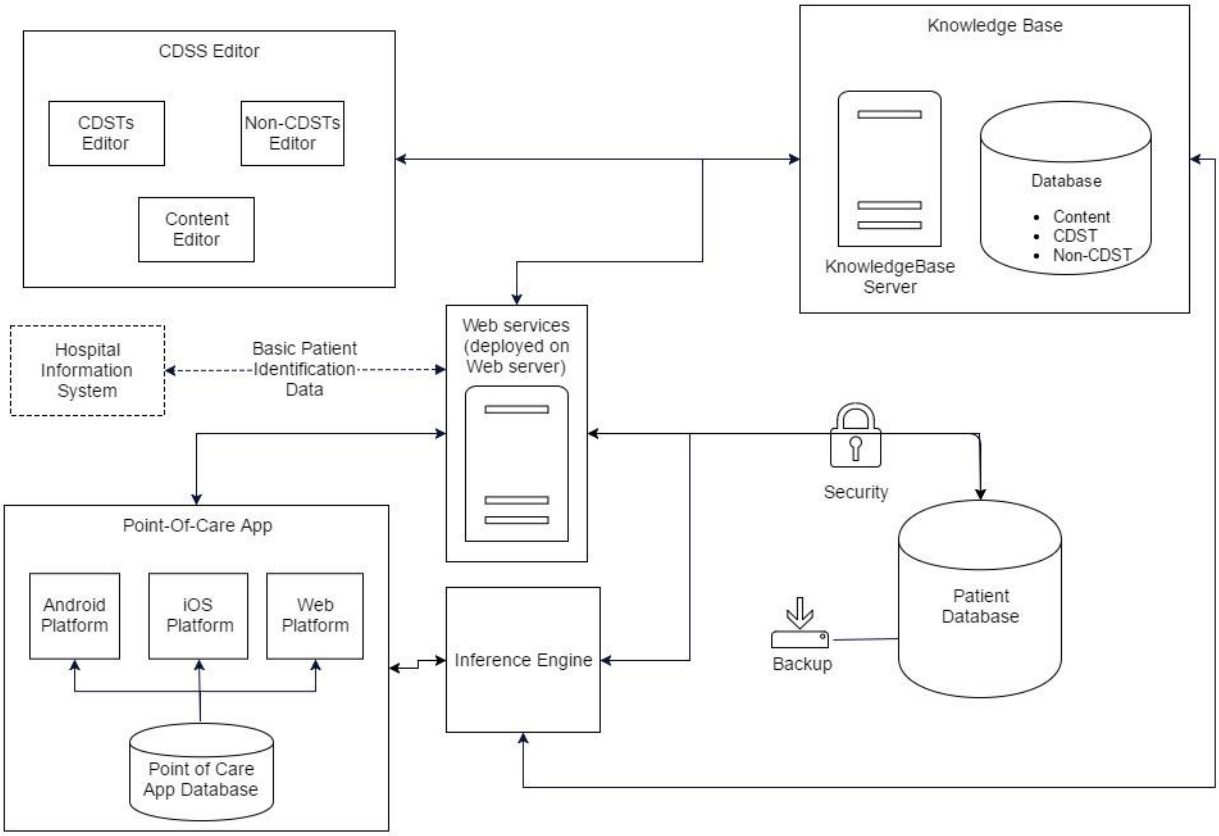

Fig 8: System Architecture and Main Components of ZAZA Pro

ZAZA Pro consists of the below main components:

1. **Knowledge Base Server** for saving & delivering Content, CDST and Non-CDST
    a. **Server** for delivery & communication
    b. **CDST, Content and Non-CDSTs Databases** for storing relational data
2. **Point-Of-Care App**
    a. iOS App
    b. Android App
    c. Web Access
3. **CDSS Editor –** Provides web interface for creation, verification and validation of CDSTs and Non-CDSTs
4. **CDSTs and Non-CDSTs –** running on Point-Of-Care apps
5. **Web Services** to transfer data between the components

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6. **Patient Database –** Independent secured database to store patient data
7. **Inference Engine –** Processing CDSS clinical decision rules
8. **Interface to Hospital Information System –** To obtain basic & necessary patient data from HIS like, patient identifier, name, hospital number, etc.
9. **Other Components –** Backup and Security of Patient Database

3.4.2 Server Components

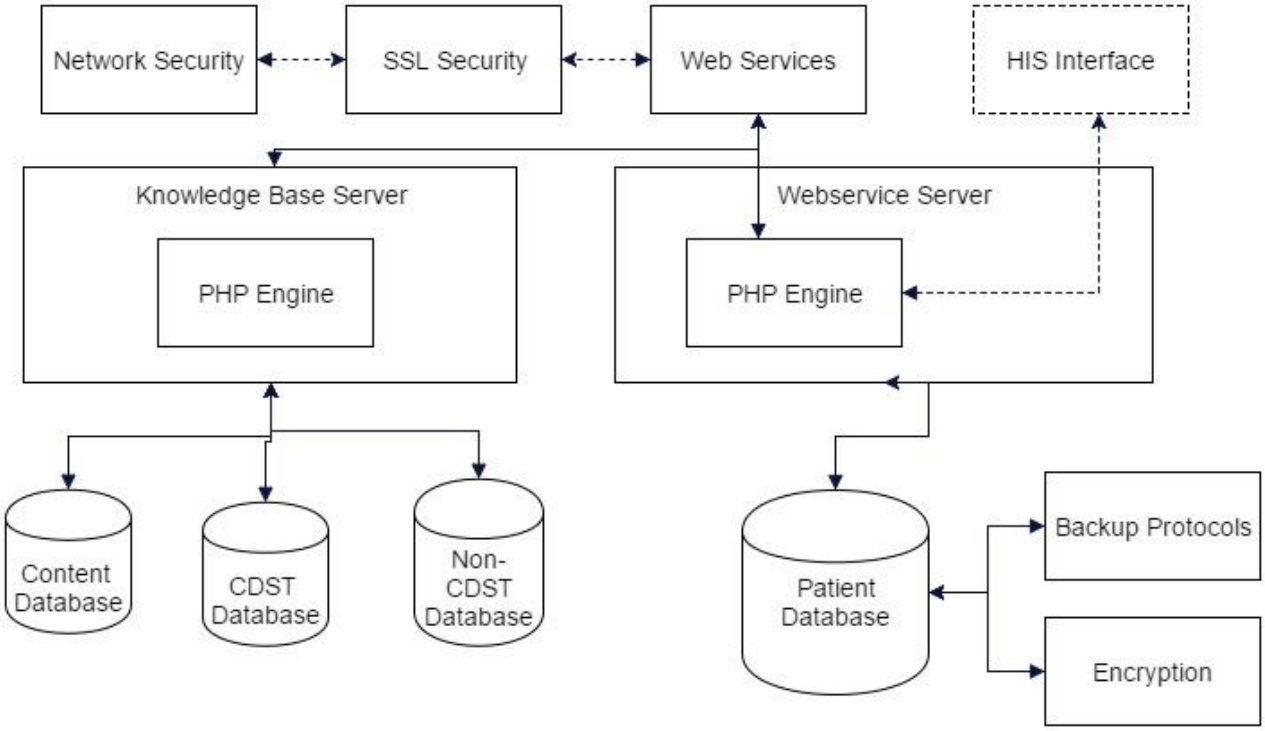

Fig 9: Structure and Server Components of ZAZA Pro

Server Components

- **Knowledge Base Server –** Processor for Content, CDSTs and Non-CDSTs
- **Web Services –** Transfer of data between different components
- **SSL Security –** Implement Transport Layer Security
- **Network Security –** Firewalls, Network access protocols etc.
- **PHP Engine –** To deploy backend code and web services
- **Backup Protocols –** Module for Backing up and Restoration of data
- **Encryption –** Component for Encrypting Data and Databases & Key management

Databases

- **Patient Database –** Store Patient Data
- **Content Database –** Store the Medical Content of the Knowledge Base
- **CDST Database –** Store all the CDST data
- **Non CDST Database –** Store all the Non-CDST data

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3.4.3 Client Components

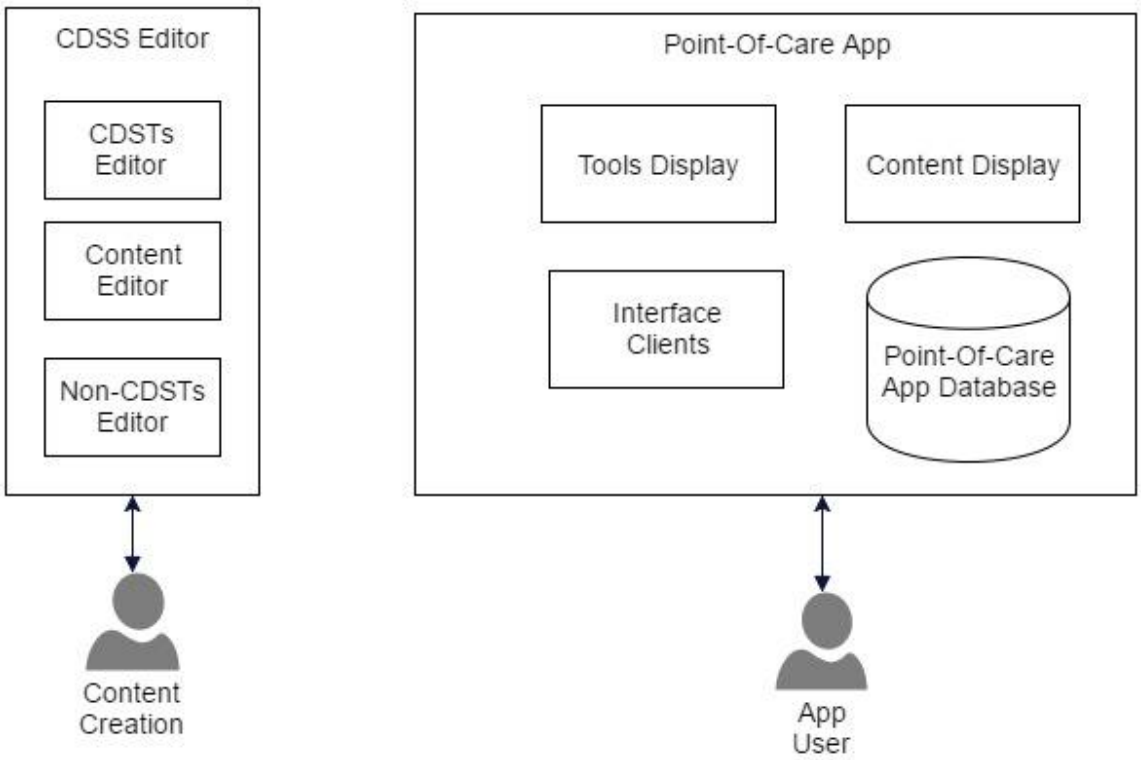

Fig 10: Structure and Clients Components of ZAZA Pro

The Client Side consists of:

1. **Point of Care App**
   a. **Tools Display –** Engine and Interface to display interactive tools
   b. **Content Display –** Engine and Interface to display content
   c. **Interface Client –** Communication client to the server using webservice components
   d. **Point-Of-Care App Database –** Storage of Tools, Content in the App
2. **CDSS Editor**
   a. **CDST Editor –** For creation and management of CDST tools
   b. **Content Editor –** For creation and management of Content
   c. **Non-CDSTs Editor –** For creation and management of Non-CDSTs Editor

3.4.4 Environmental Conditions

The system can be used under normal hospital operating conditions and in clinics for private practice.

3.5 Product Labelling

As a standalone software system also, product labelling is provided via software interface as follows:

1. **Use instruction**, such as "**How to use**" a CDS Tool including **used symbols** are provided within the "How to" section of the tool available in side bar of each CDS Tool.

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2. **Manufacturer Information and product Information**, such as version, legal manufacturer, classification and legal manufacturer and CE mark is provided within "**Imprint**" section available in the side bar of each CDS Tool.
3. Further **regulatory information and product definitions** are defined and provided in **tool profiles** available via side bar within each CDS Tool.

Please see screenshots documentation in chapter 6.4 including screenshots showing the buttons to enter "How to Use", "Imprint" or "Tool profile" Screens considered the product labelling or user manual.

4 Product Classification

4.1 Classification according to Medical Device Directive 93/42/ECC

The product is classified as Class IIa medical device in accordance to a general risk based approach considering possible non-serious harms.

The risk based approach was selected since none of the rules 9-11 for active medical devices applies and rule 12, resulting in class I, deemed not appropriate.

4.1.1 Classification Justification

ZAZA Pro, as standalone software system, is considered as an active medical device, even so it is not influencing of controlling any active medical device. The device is **classified as class IIa according to the following risk based classification:**

- No Harm possible = Class I
- Non-Serious Harm possible = Class IIa
- Serious Harm or Death possible = Class IIb

This risk based classification is used, since application of the rules from the medical device directive 93/42/EEC would result in Class I (no rule applies), as discussed below. This classification does not seem to be appropriate, compared to the rules of the risk based classification above, leading to either IIa or IIb.

As also discussed during software safety classification, the system is used to support the execution of clinical processes with no direct possibility to harm. There is always physicians or other health care professionals involved between a potential malfunction of the system and an according activity of the user. The system is used in clinical environment ensuring safety as a basic concept. Critical activities, as an example, are only done in intensive care environment. Critical medication is provided under monitoring. All these safety measures of the clinical routine are applicable and allow a classification as class IIa.

The implementing rules of directive 93/42/EEC, Annex II, IX, especially implementing rule chapter 2.3 is not applicable. The potential use of a medical device is neither controlled nor influenced by ZAZA Pro, but is under the responsibility of the executing person and the manufacturers responsibility for safe and effective use.

The classification rules of directive 93/42/EEC, Annex III, IX, have been evaluated with particular focus on chapter 3. rule 9-12, additional rules applicable for active devices:

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- **Rule 9**
  Rule 9 is not applicable, since the device is not a therapeutic device administering or exchanging energy.
- **Rule 10**
  Rule 10 is not applicable, as none of the below three applicable intended used applies:
  - supply energy, to image in vivo distributions of radiopharmaceuticals, or
  - direct diagnosis, or
  - monitoring of vital physiological processes

  As explained in chapter 2, no direct diagnosis is provided, but clinical decision support.
- **Rule 11**
  Rule 11 is not applicable, since the device is not intended to administer or remove medicine or substances to or from the body.
- **Rule 12**
  Rule 12 is applicable, but deemed not appropriate.

4.2 Software Safety Classification according to IEC 62304:2006

The software system in classified as Class B software.

4.2.1 Justification

The multidisciplinary indications of use and the accordingly used worst case approach would lead to class C, since severe harm is theoretically possible, if probability of occurrence is considered 100%. In accordance to the IEC 62304:2006, we reduce the safety class to B. The system is used to support the execution of clinical processes with no direct possibility to harm. There is always physicians or other health care professionals involved between a potential malfunction of the system and an according activity of the user. The system is used in clinical environment ensuring safety as a basic concept. Critical activities, as an example, are only done in intensive care environment. Critical medication is provided under monitoring. All these safety measures of the clinical routine are considered external safety measures, that allow reduction of safety class from C to B.

Furthermore, the system cannot be used without prior validation of the supported clinical processes performed and confirmed by the responsible CMS user. The mobile app client software will not work with not-validated clinical process content.

4.3 Classification according to NBOG BPG 2009-3

ZAZA Pro is considered as:

- MD 1111 Software

4.3.1 Classification Justification

There is not classification, which is addressing ZAZA Pro in more detail. Therefore, only MD1111 Software is applicable.

4.4 Classification according to UMDNS

According to the Universal Medical Device Nomenclature System (UMDNS), ZAZA Pro is considered as:

- 17-762 Data-Management-System

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4.4.1 Classification Justification

The intended use of our device is not perfectly met by data management system. Nevertheless, it is the closest ID, since ZAZA Pro is managing data, data being understood as clinical process and medical reference data.

An alternative possible classification would be:

- 17-222 Information System

We decided to use 17-762 Data-Management-System, since we do not consider our system as a kind of clinical or hospital information system, which are commonly understood as overall information systems to manage clinics, hospitals, private practices or particular departments as a whole. ZAZA Pro may be used together with an according information system, but is not considered an information system, in this sense.

5 Product Variants, Compatible Devices and Accessories

5.1 Product Variants

The product will be launched for smartphone and tablet mobile devices with a smartphone and a tablet variant.

5.2 Compatible Devices

5.2.1 IOS Devices

1. iPod Touch 5 Generation
2. iPhone 4S
3. iPhone SE
4. iPhone 5
5. iPhone 5S
6. iPhone 5C
7. iPhone 6
8. iPhone 6 Plus
9. iPad 2 and higher
10. iPad Airs
11. iPad Minis
12. iPhone 7
13. iPhone 7 Plus
14. iPad Pro

5.2.2 Android Devices

Wide range of phones that comprise of these companies and specifications

1. Google Nexus
2. Samsung
3. HTC
4. Motorola
5. LG
6. Asus
7. Lenovo
8. Huawei

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5.2.3 Specifications

Parameters	Values
Minimum RAM	1 GB Recommended
Screen sizes	Small, normal, large, x-large
Minimum internal storage size	1 GB
Minimum external storage size	1 GB
Data networks	2G, 3G, Wi-Fi

5.3 Accessories

N/A

6 Appendix

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6.3 Glossary

6.3.1 General Terms

No.	Terms / Short-form	Definition
1	Activity	Action; basic dimension of process management
2	Clinical Condition	Particular state of health of a person
3	CDSS	Clinical decision support system
4	CDSTs	Clinical decision support tools
5	Clinical Guideline	Descriptions of how to act in a certain clinical condition
6	Clinical Pathway	Structured multidisciplinary management plan with clinical interventions details
7	Clinical Process Management	Identification, configuration, documentation, implementation, controlling and improvement of clinical processes of medical activities

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8	CMS	Content management system; an application or software that facilitates the entry, modification and retrieval of digital content
9	CPP	Clinical Process Protocol; document that describes a clinical process (chronological, logical sequence of tasks), including SOPs (detailed written instructions of performing activities) and additional information
10	Decision	Making a choice out of a number of different options after consideration of environmental conditions
11	Effectiveness	Capability of producing a desired result (refers to the degree to which a planned effect is achieved)
12	Efficiency	Capability of achieving a result with the least effort (a measure of cost effectiveness, effort)
13	Effort	Amount of energy or work to fulfill an activity or to achieve a result
14	EHR / EMR	Electronic health record / Electronic medical record
15	Entity	A product, process, activity, machine, service, system, department, company, institution or organization
16	Management	Activities and methods used to coordinate, direct and control organizations
17	POC	Point-of-care
18	Process	Chronological logical sequence of activities/tasks to produce a result
19	Procedure	A particular method of performing a task or a process
20	Quality Management	Activities that management carry out in an effort to implement the quality policy
21	Quality Policy	Describes an organization's commitment to quality (quality policy statement)
22	Record	Document of what has happened and the result, reflects how well activities are performed or results are achieved
23	Result	Final consequence (effect, outcome) of an activity or a process
24	Strategy	Plan for achieving an objective /desired results
25	SOP	Standard operating procedure; detailed instruction of performing a task to achieve uniformity in results
26	Task	Activity that needs to be fulfilled within a certain time (deadline)

6.3.2 Mathematical Concepts Used in Clinical Decision Support

Set theory: a branch of mathematical logic that studies sets. A **set** is a well-defined collection of unique objects or elements (e.g. a set of polygons). Each **element** of the set is distinguishable from the others. A **subset** is any collection of elements of a known set. The basic **set operations** are intersection and union. The **intersection** (Schnittmenge) of two sets is the set of elements, which both sets have in common. The **union** (Vereinigungsmenge) of two sets is the set of all elements that belong to either set. Sets are used to **formalize logical operations** in a way that machines (computers) can understand.

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Boolean algebra: a branch of algebra. Compared to the well-known elementary algebra, where the values of variables are numbers, in logic algebra **or Boolean algebra** the values of variables are true and false (denoted 1 and 0). Where the main operations in elementary algebra are addition and multiplication, the main operations of Boolean algebra are **conjunction** and denoted as ∧, **disjunction** or denoted as ∨, and negation not denoted as ¬. **Logical conjunction** is the operation on two logical values, typically the values of two propositions, that produces a value of true if and only if both of its operands are true. **Logical disjunction** is an operation on two logical values, typically the values of two propositions, that has a value of false if and only if both of its operands are false. Logic algebra is a formalism for describing logical relations in the same way that ordinary algebra describes numeric relations. Boolean algebra has been fundamental in the development of digital electronics, and is provided for in all modern programming languages.

Probability Theory: Every day medical practice contains many examples of likelihood, as described in **probability theory**. **Probability** is represented numerically by a number between zero and one. **Conditional probability** is the probability of an event given the occurrence of another event. Computerized systems can use some numerical representation of probability in order to combine statements into conclusions.

Fuzzy set theory: membership in a set is not an all-or-nothing-phenomenon, but an element is a member of the set with a certain probability. **Fuzzy logic** is used to handle decision making problems in which uncertainty plays a role.

Decision theory: the choices of an individual and the underlying reasoning. Normative decision theory gives advice on how to make the best choices, given a set of uncertain beliefs and a set of values. A decision rule in medicine is a representation of deterministic reasoning in which branching logic is used in combination with data to reach conclusions regarding diagnosis, treatment and other important clinical goals.

6.3.3 Terms of Risk Management

No.	Terms / Short-form	Definition
1	Harm	Physical injury, compromised health
2	Hazard	Potential source of harm
3	Hazardous Situation	Location and surroundings in which one is exposed to a hazard
4	PHA	Preliminary Hazard Analysis
5	Risk	Combination of probability of harm and severity of harm (very likely to very unlikely, very harmful to slightly harmful)

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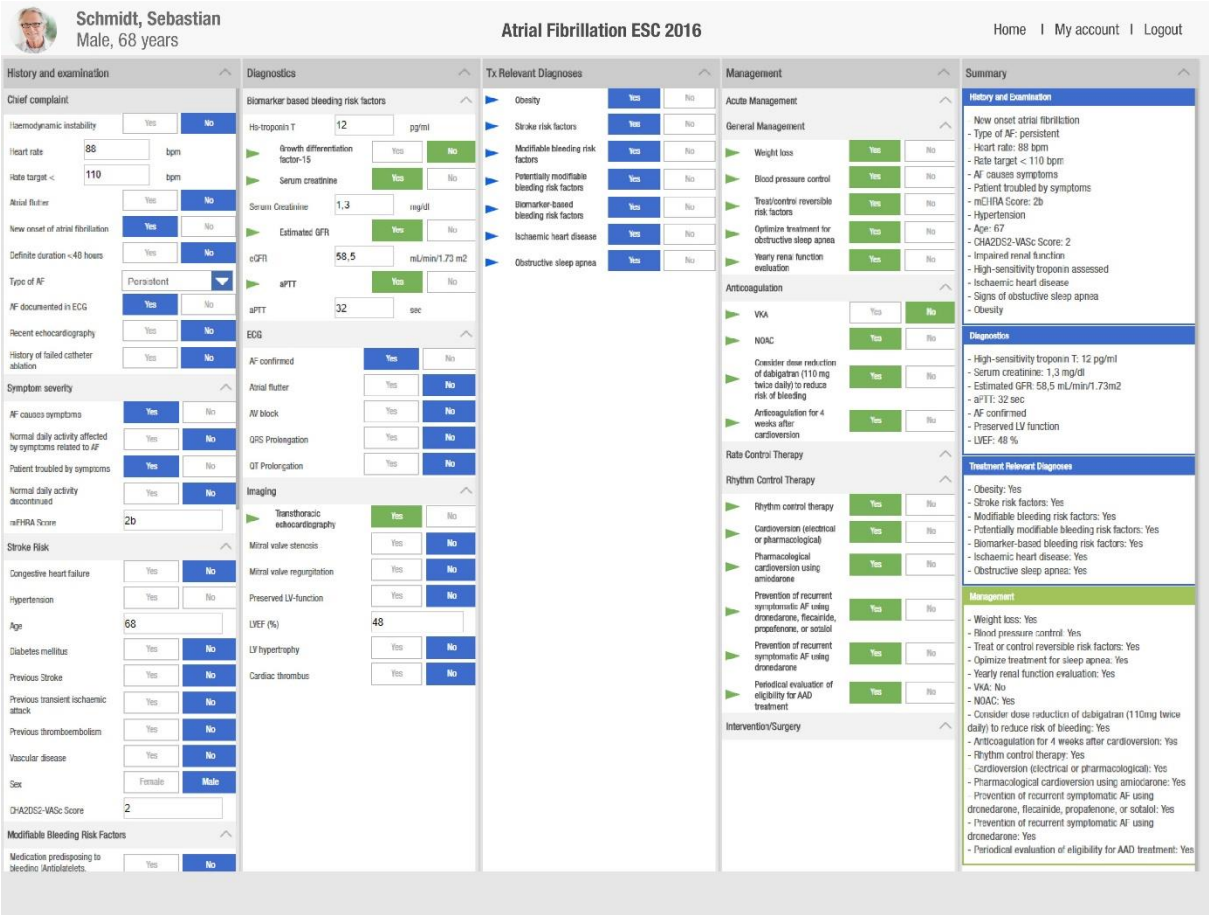
6.4 Screenshot Documentation



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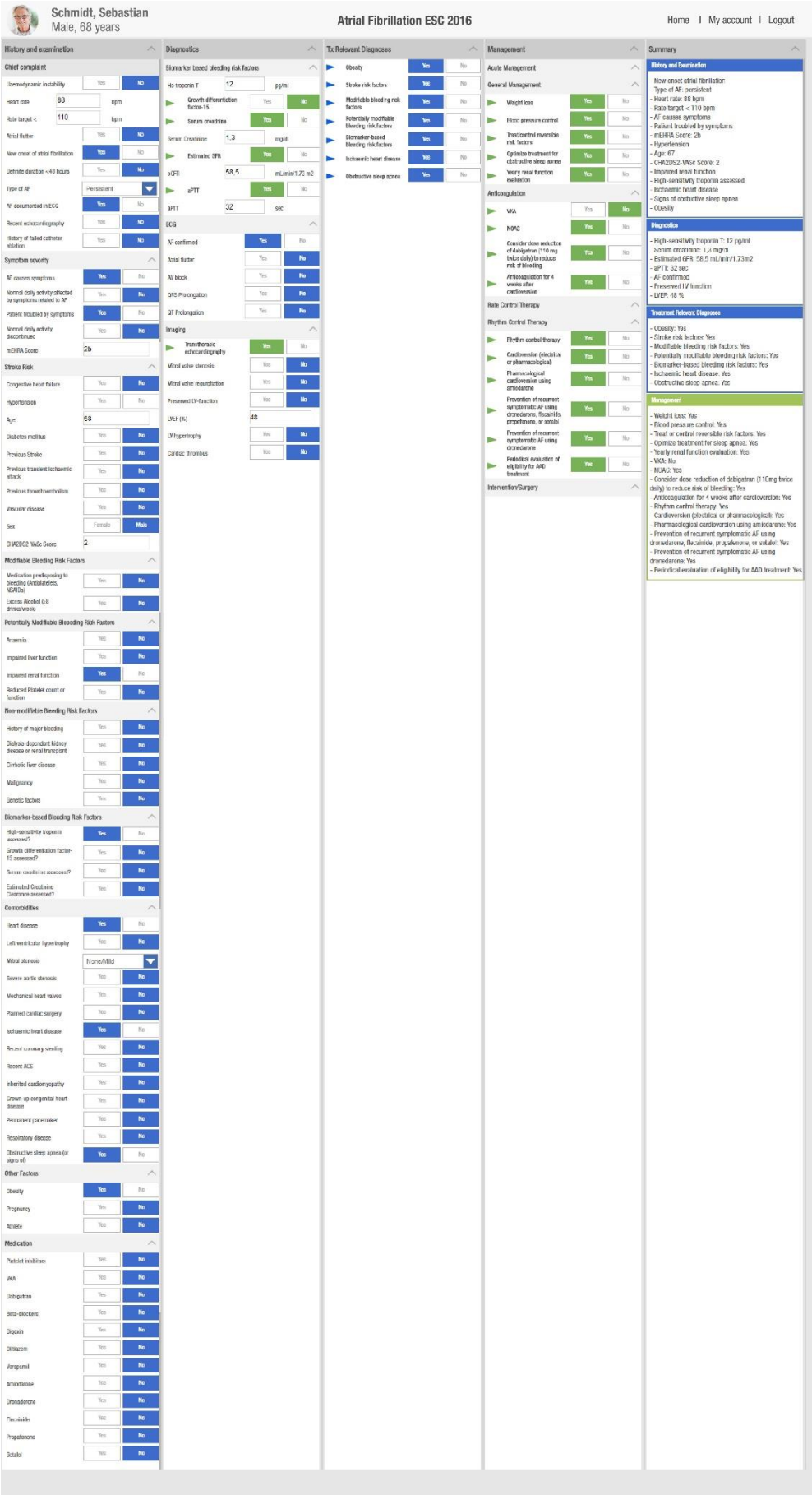

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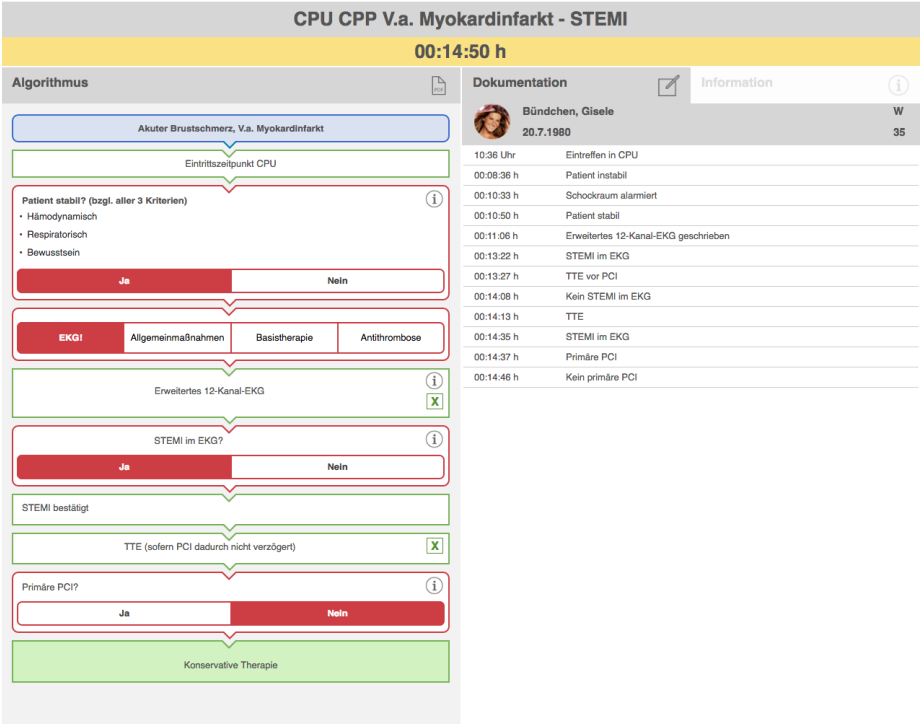

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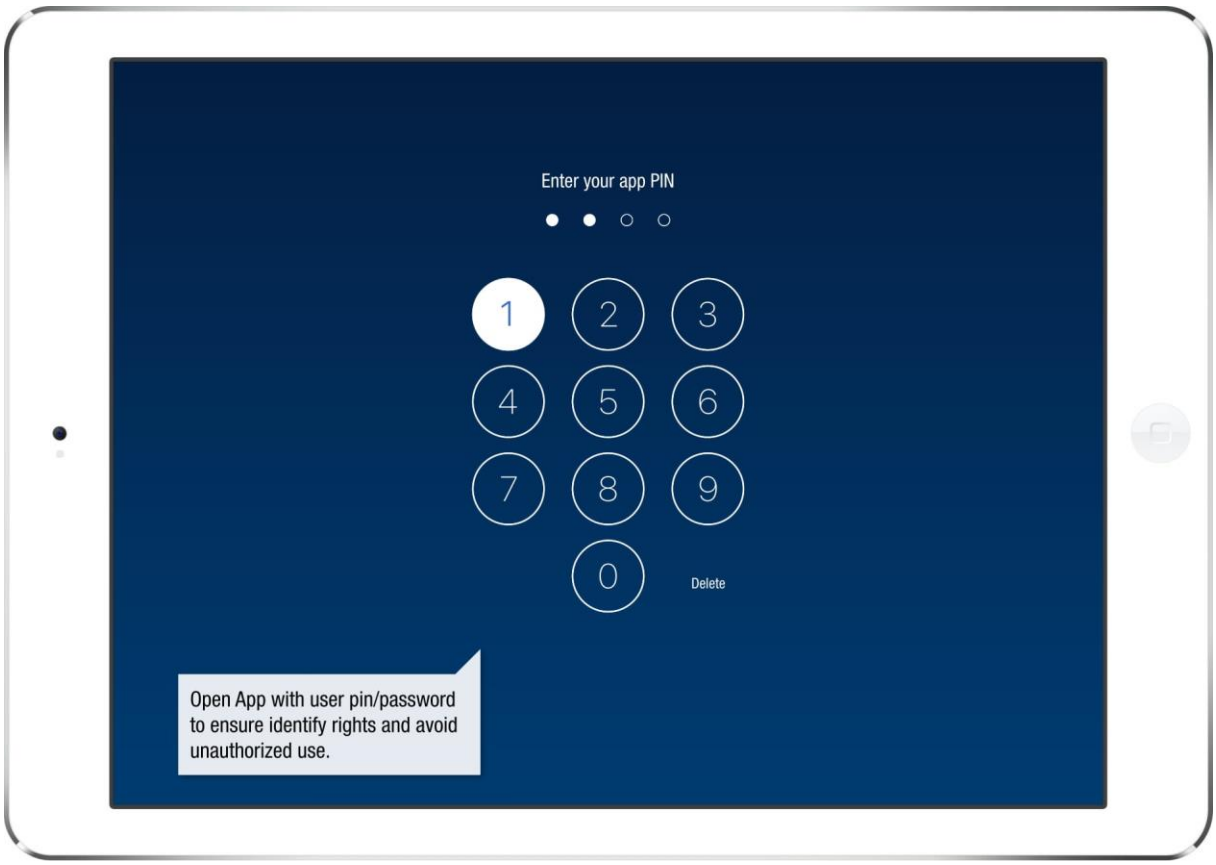

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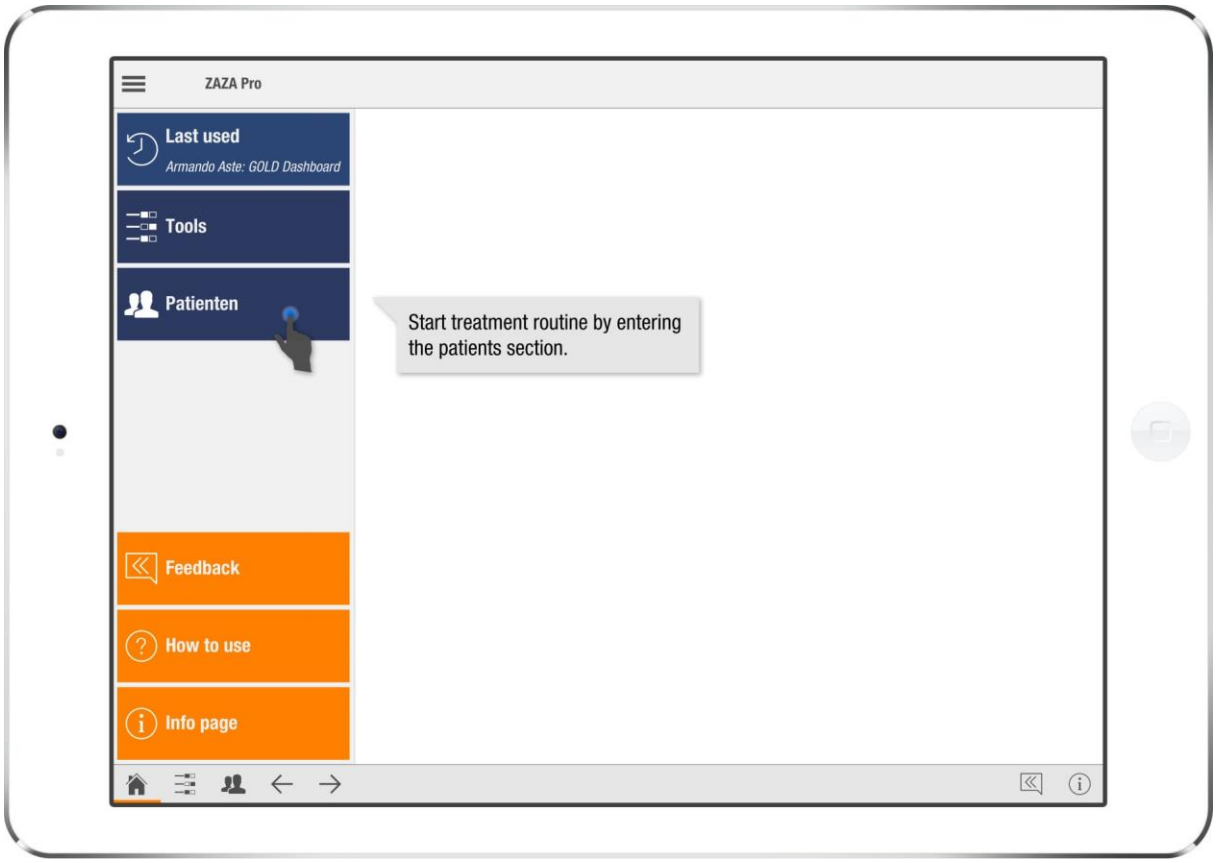

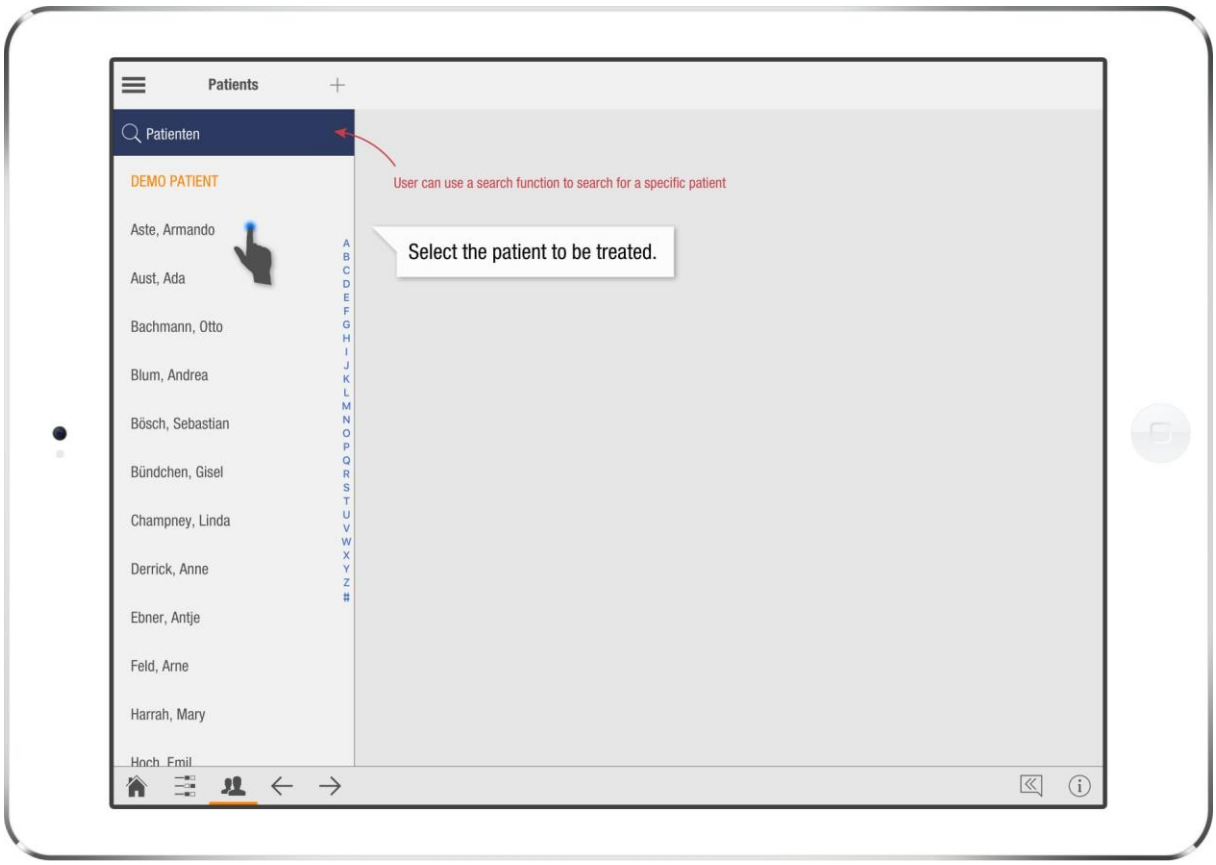

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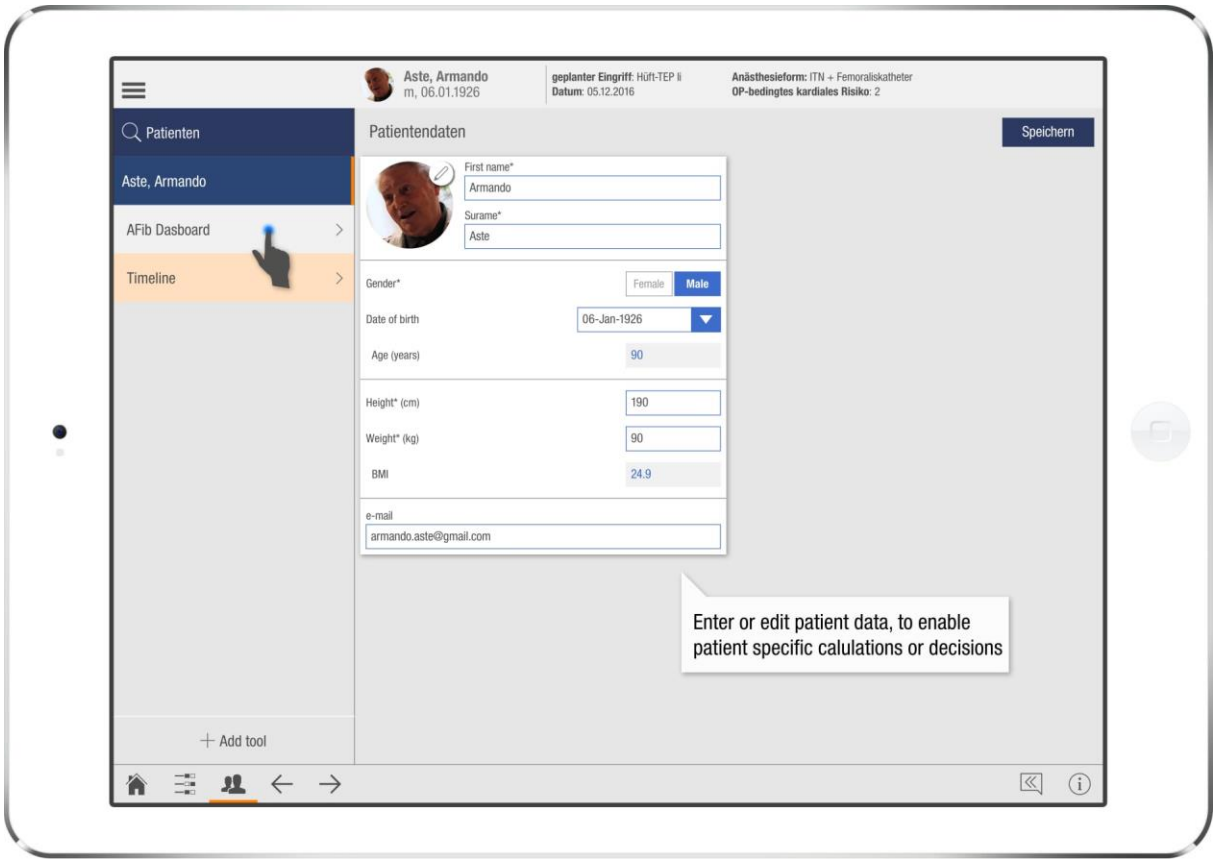

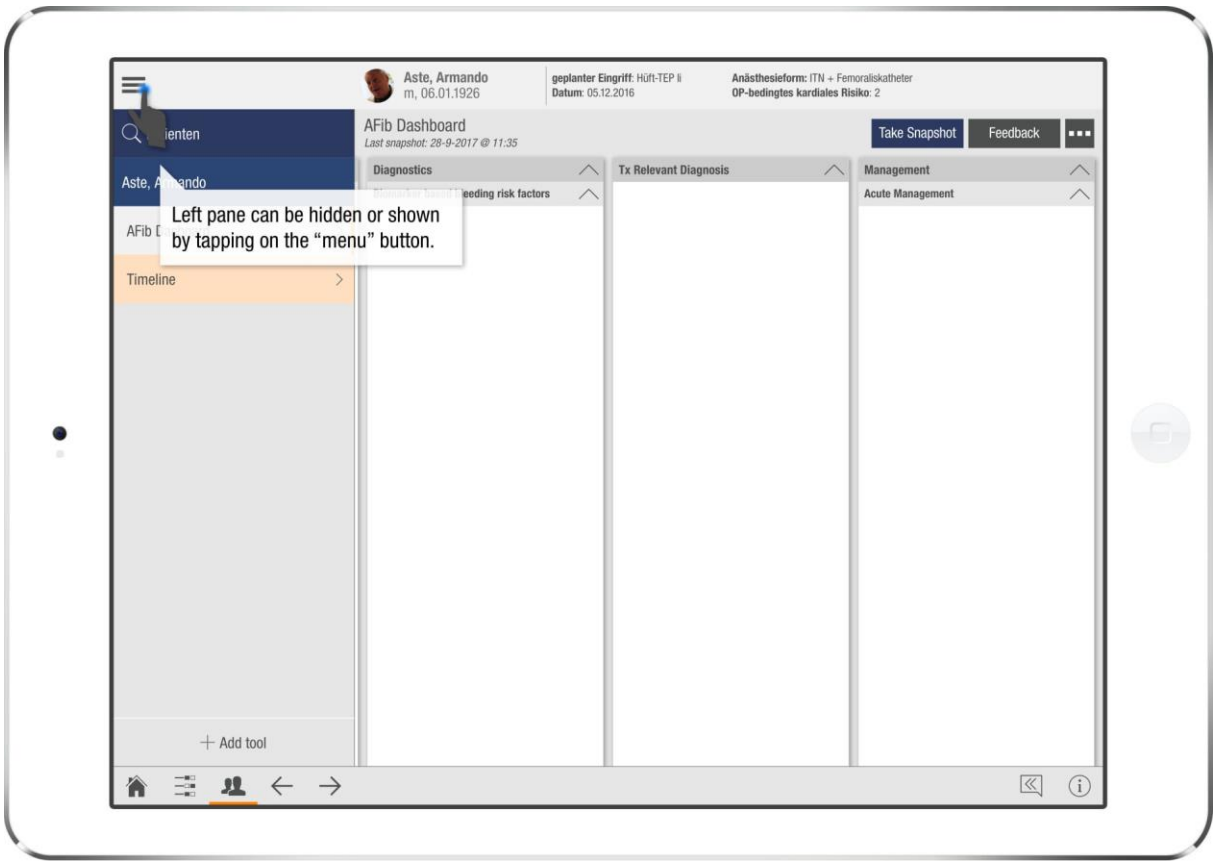

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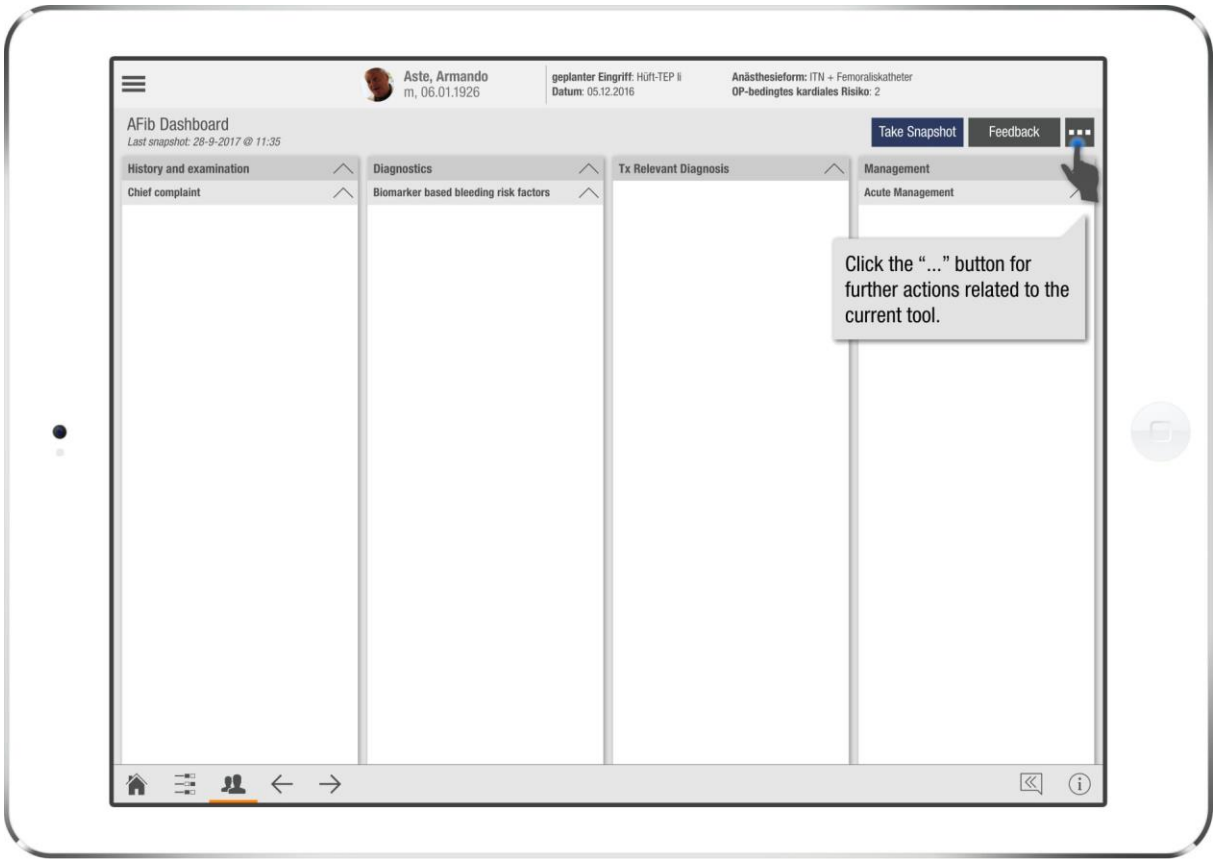

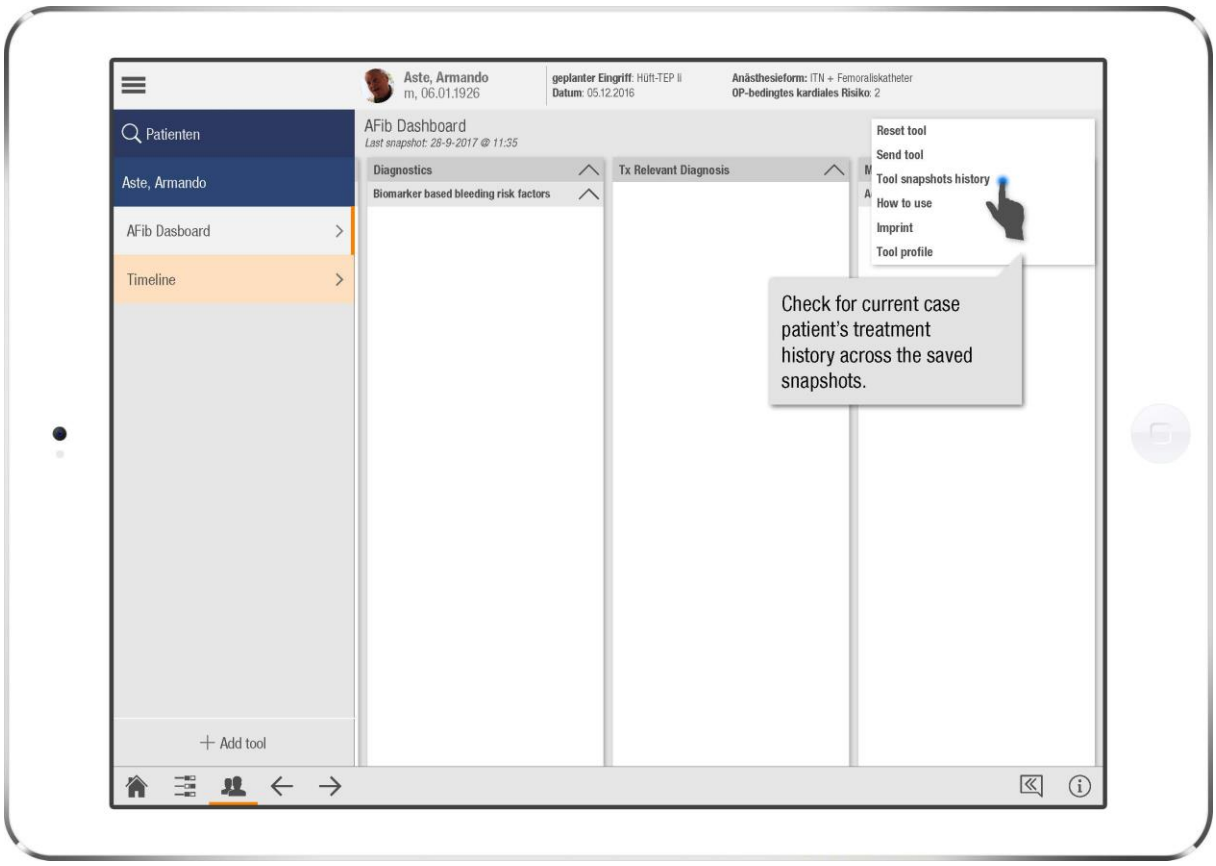

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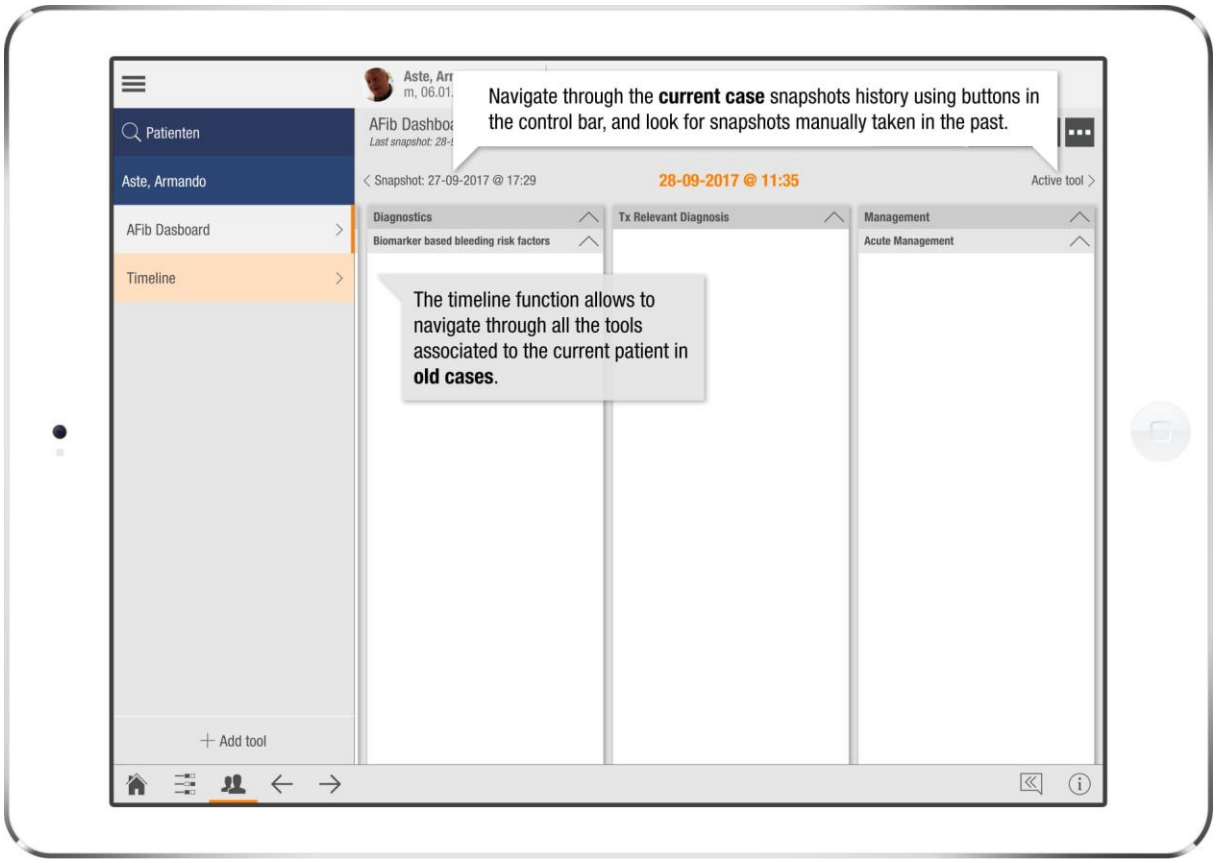

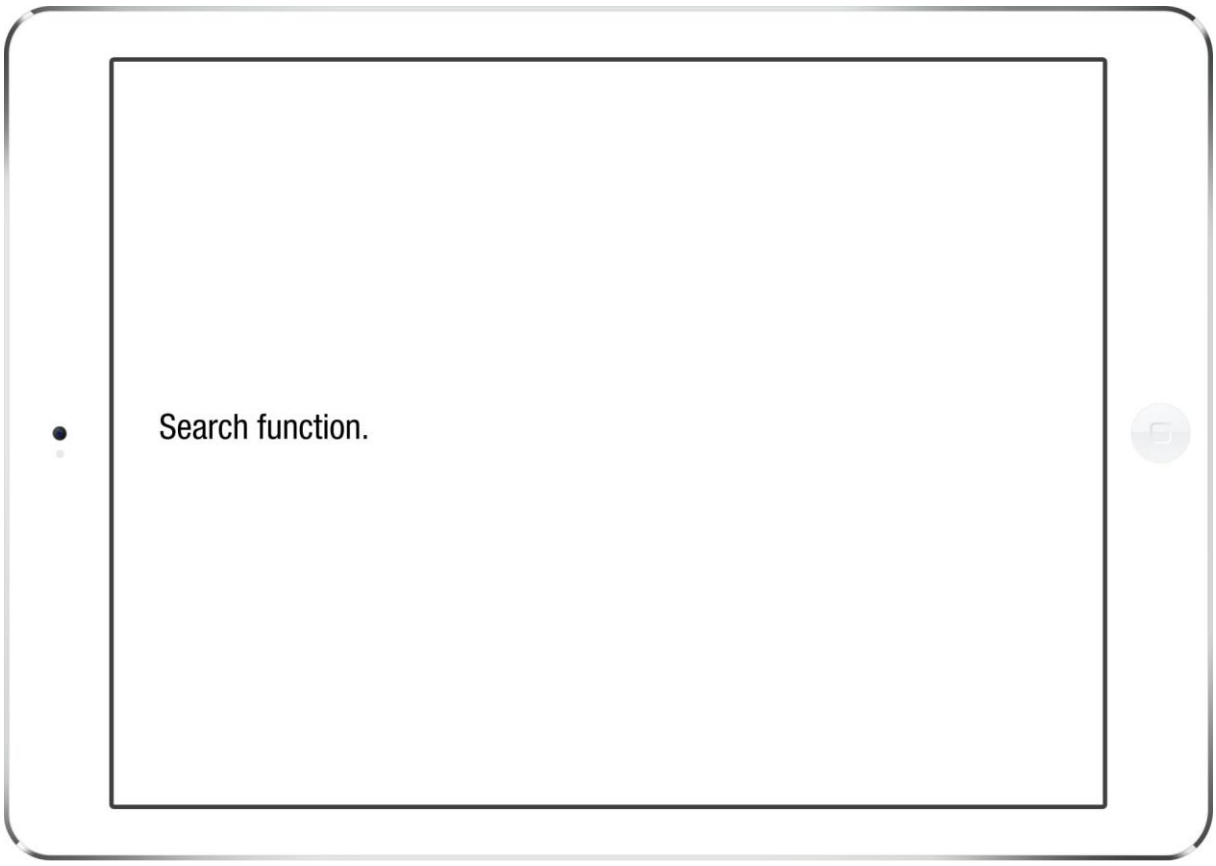

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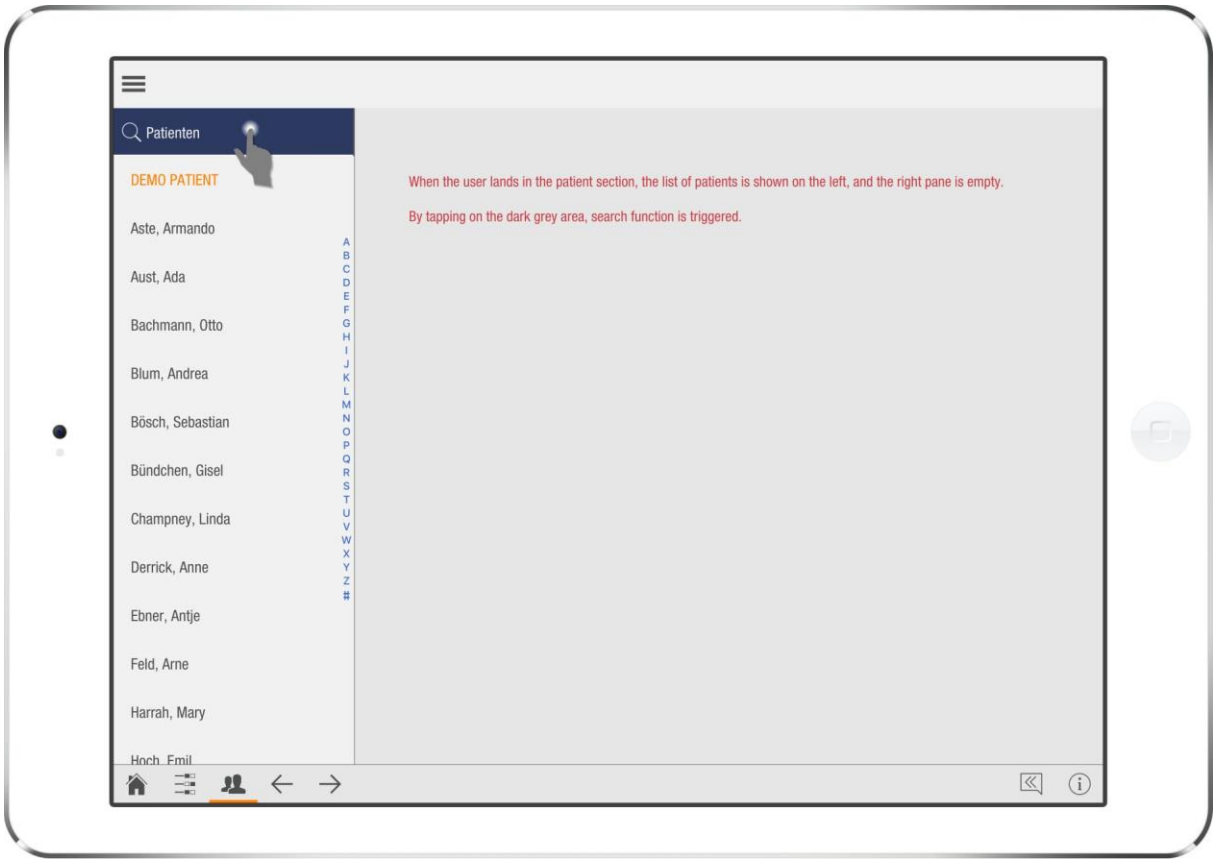

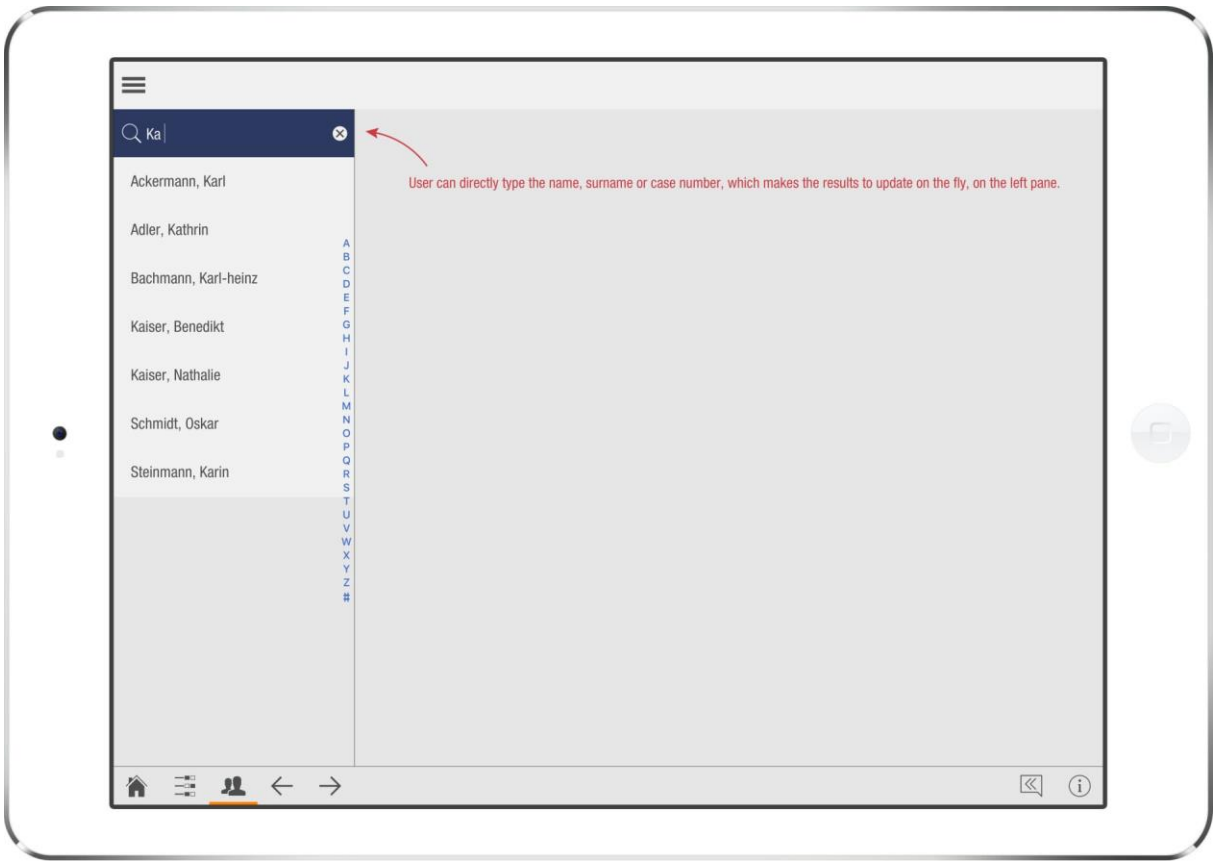

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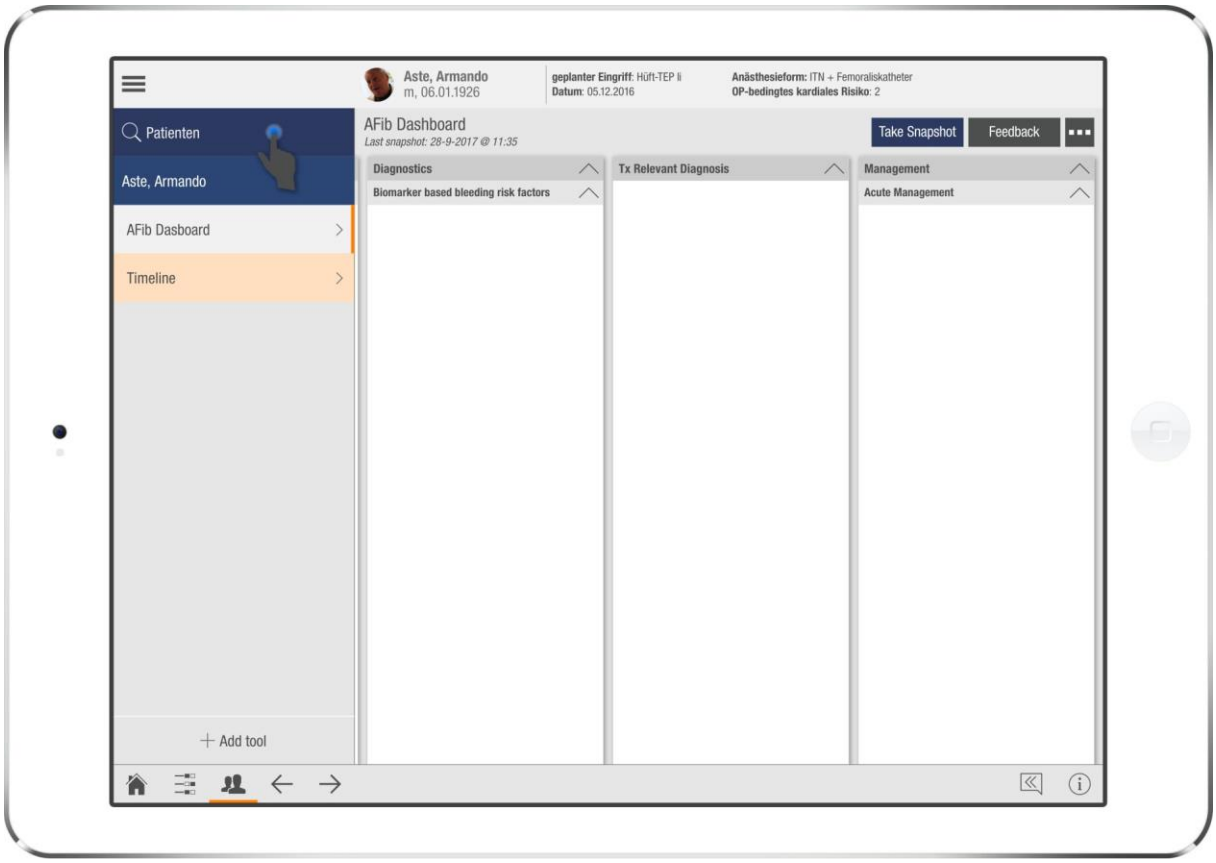

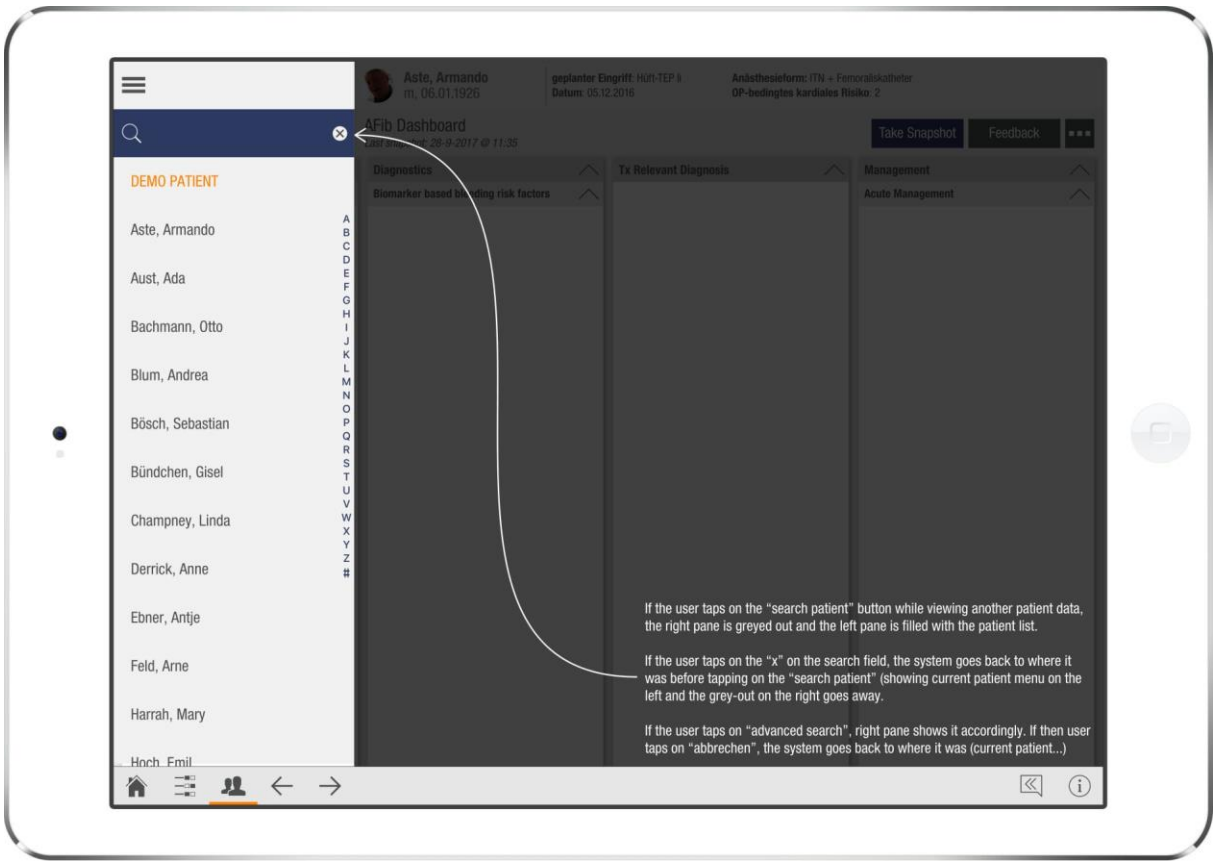

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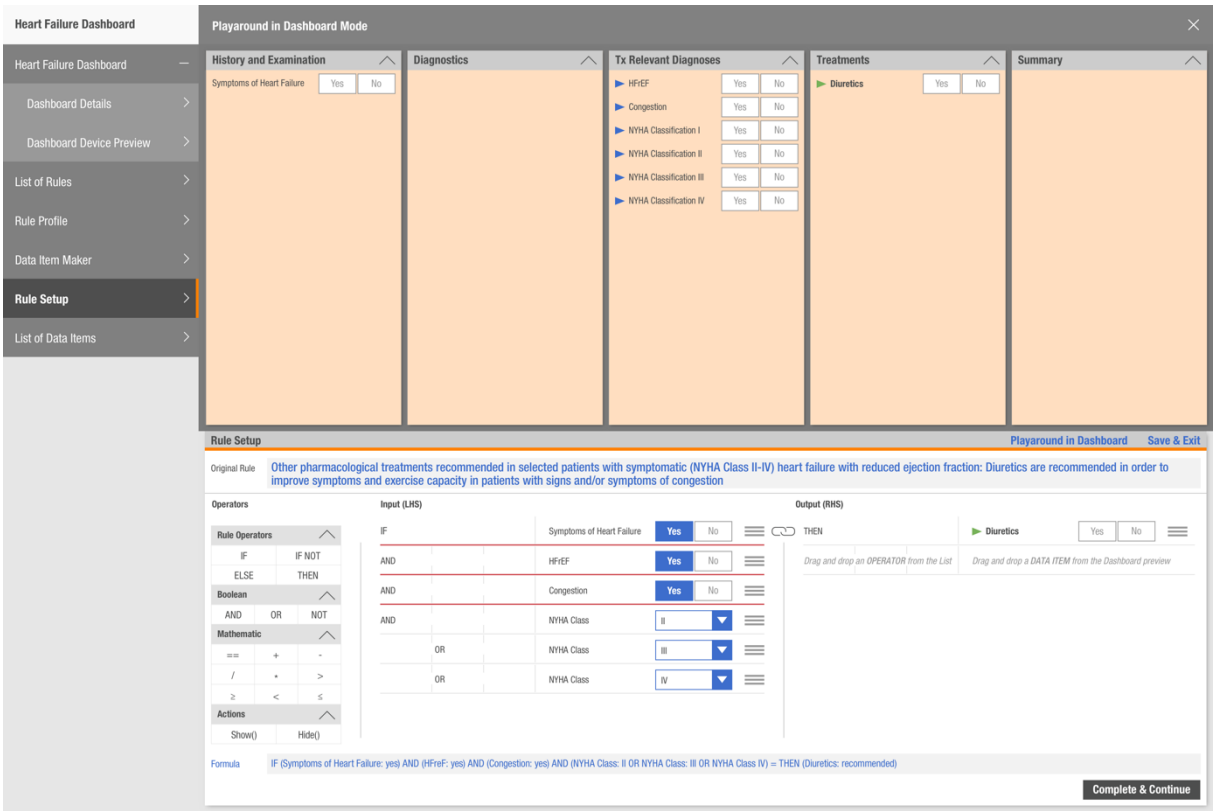

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